

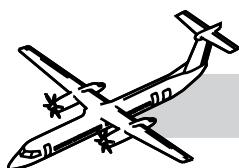
CHAPTER 26 FIRE PROTECTION

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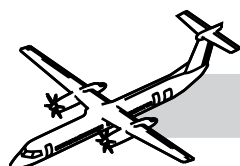


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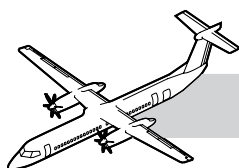
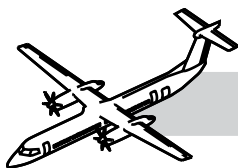


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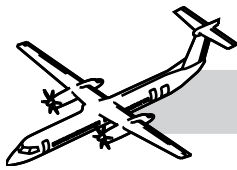
CHAPTER 26

FIRE PROTECTION



26-00-00 INTRODUCTION

The fire protection system includes components throughout the aircraft to provide detection, indication and extinguishing of fire conditions.



GENERAL

NOTES

FIRE DETECTION SYSTEM

Refer to Figure 26-1. Fire Protection Functional Block Diagram.

The fire detection system is designed to sense fire, overheat and smoke conditions.

The fire detection system consists of:

- Nacelle fire detection system
- Baggage compartment smoke detection system
- Lavatory smoke detection system
- APU fire detection system.

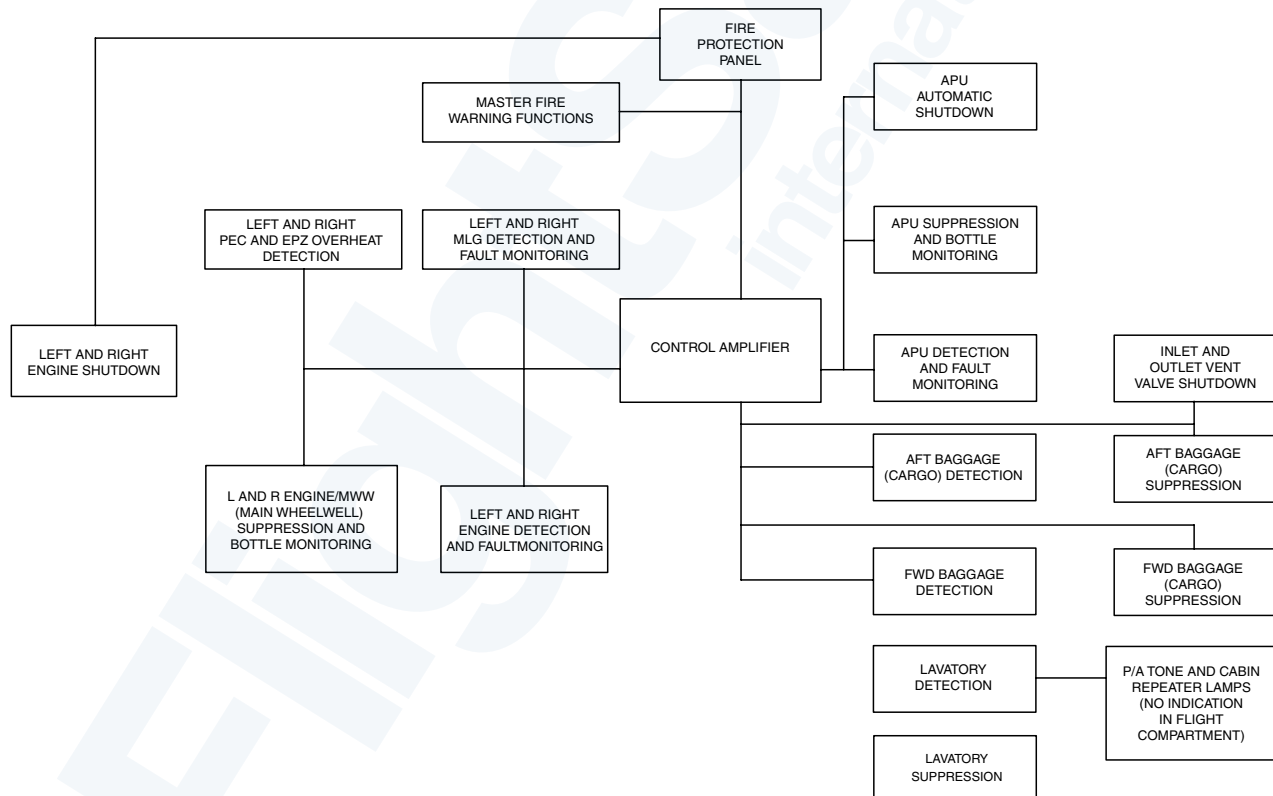
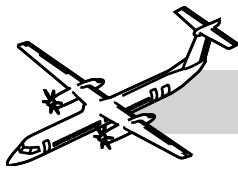


Figure 26-1. Fire Protection Functional Block Diagram



FIRE EXTINGUISHING SYSTEM

Refer to Figure 26-2. Fire Protection Panel (FPP).

The fire extinguishing system provides components throughout the aircraft for the extinguishing of detected fires.

The fire extinguishing system consists of:

- Nacelle fire extinguishing system
- Baggage compartment fire extinguishing system
- APU fire extinguishing system
- Lavatory fire extinguishing system
- Portable hand-operated fire extinguishers.

NOTE

Fire or smoke detection is shown on the FPP, CAWP, and Glareshield Panel. Indication is provided at the cabin repeater lights for the lavatory smoke detection.

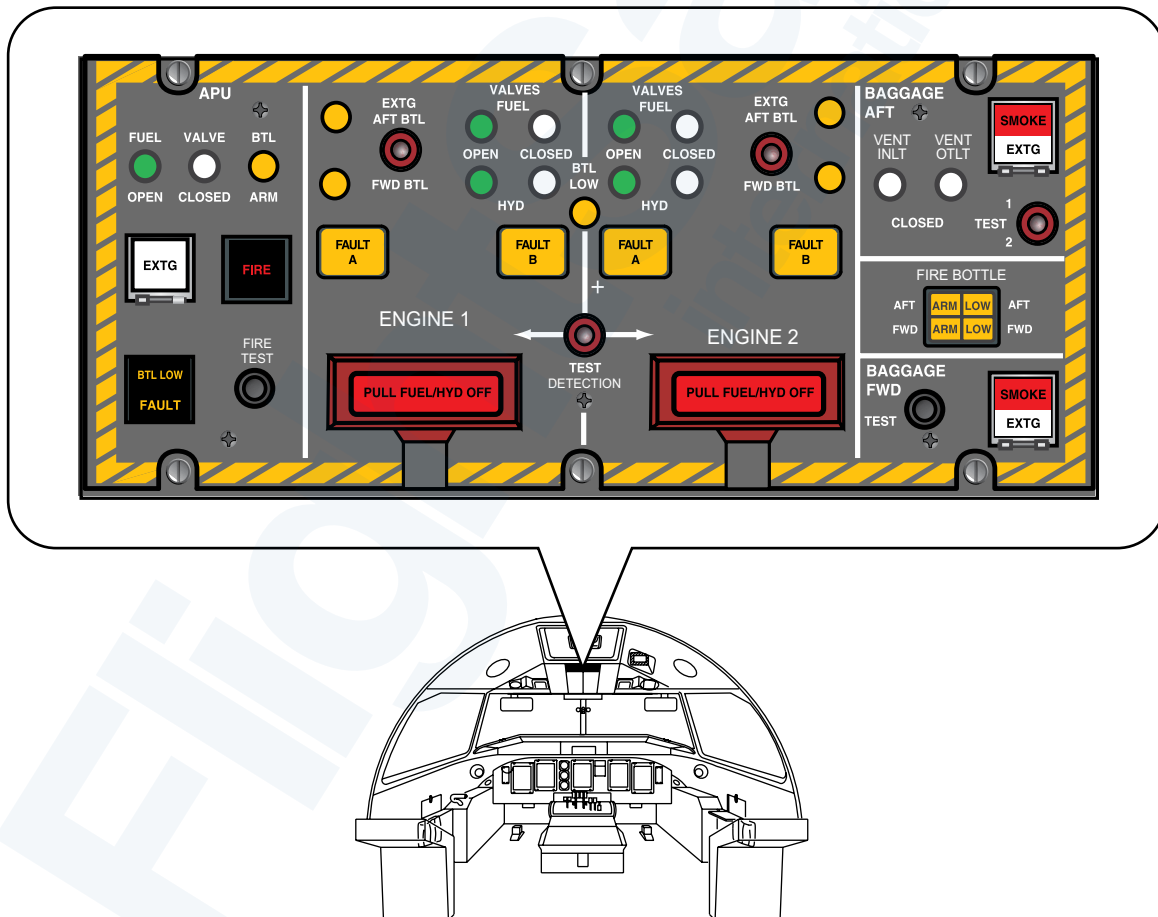


Figure 26-2. Fire Protection Panel (FPP)

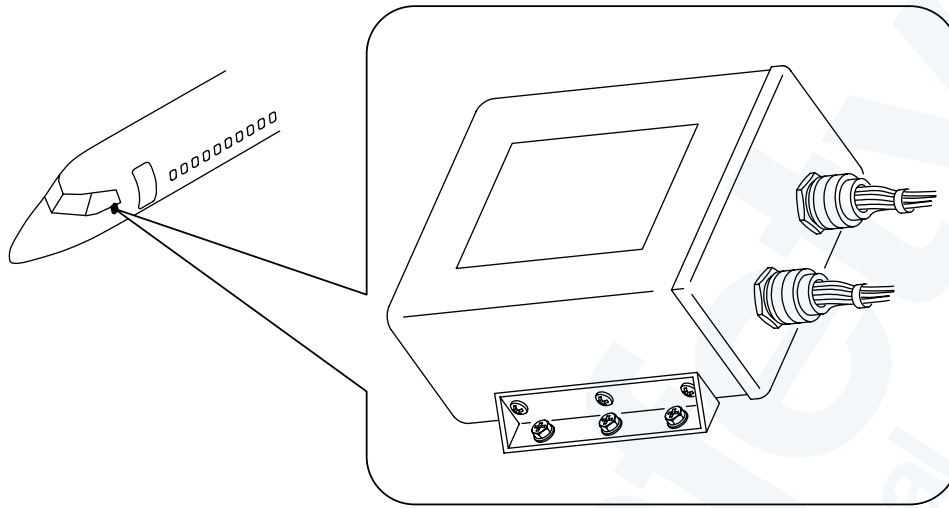
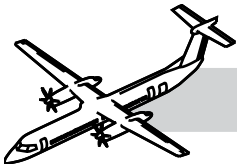


Figure 26-3. Control Amplifier

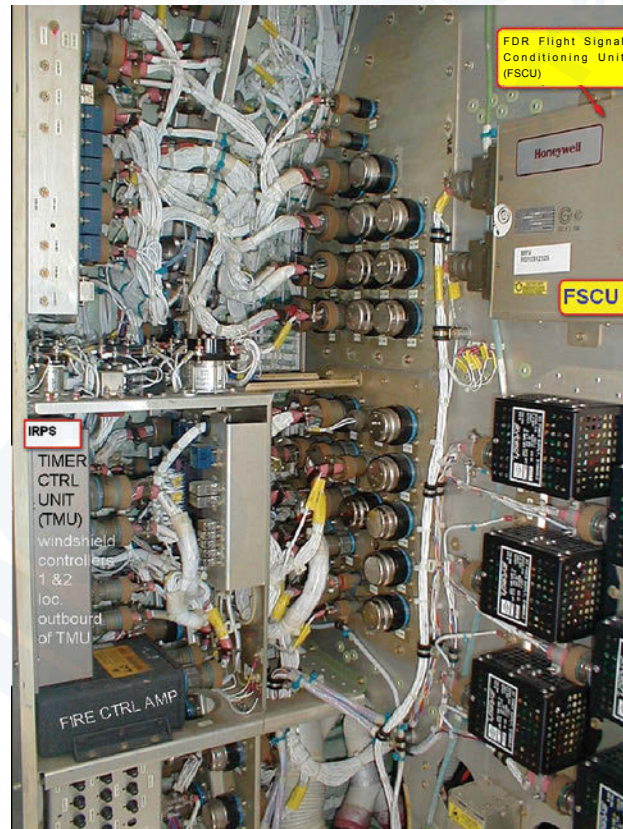
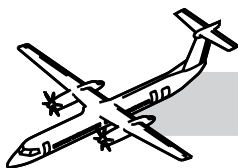


Figure 26-4. Control Amplifier



COMPONENT DESCRIPTION

Control Amplifier

Refer to:

- Figure 26-3. Control Amplifier.
- Figure 26-4. Control Amplifier.

[Question: Where is control amplifier located?](#)

The control amplifier is behind the forward wardrobe on the No. 2 equipment panel.

The control amplifier is an electronic unit. BIT control and fault monitoring are accomplished through the control amplifier. There is redundancy in the design of the unit to make sure that there is maximum functional reliability.

The primary functions of the fire protection system control amplifier for the Engines, APU and Baggage compartments are:

- Fire or smoke detection monitoring status
- Bottle pressure status
- Squib continuity status
- Fault status
- Arm status
- Test functions and indicated results
- Supply signal to the warning lights (CAWP)
- Control the aural alerts.

NOTE

Loss of control amplifier will not cause loss of detection or suppression capability in the fire protection system of the aircraft.

[Question: What if control amplifier losses?](#)

Fire Protection Panel (FPP)

Refer to Figure 26-2. Fire Protection Panel (FPP).

The fire protection panel, installed in the flight compartment on the overhead console, provides:

- Visual indication of fire
- Overheat conditions in the engines APU fire zones and
- Smoke in the forward and aft baggage compartments.

The face of the FPP is divided into the following three areas:

- Engine fire protection panel (Center)
- Baggage compartment fire protection panel (Right)
- APU fire protection panel (Left).

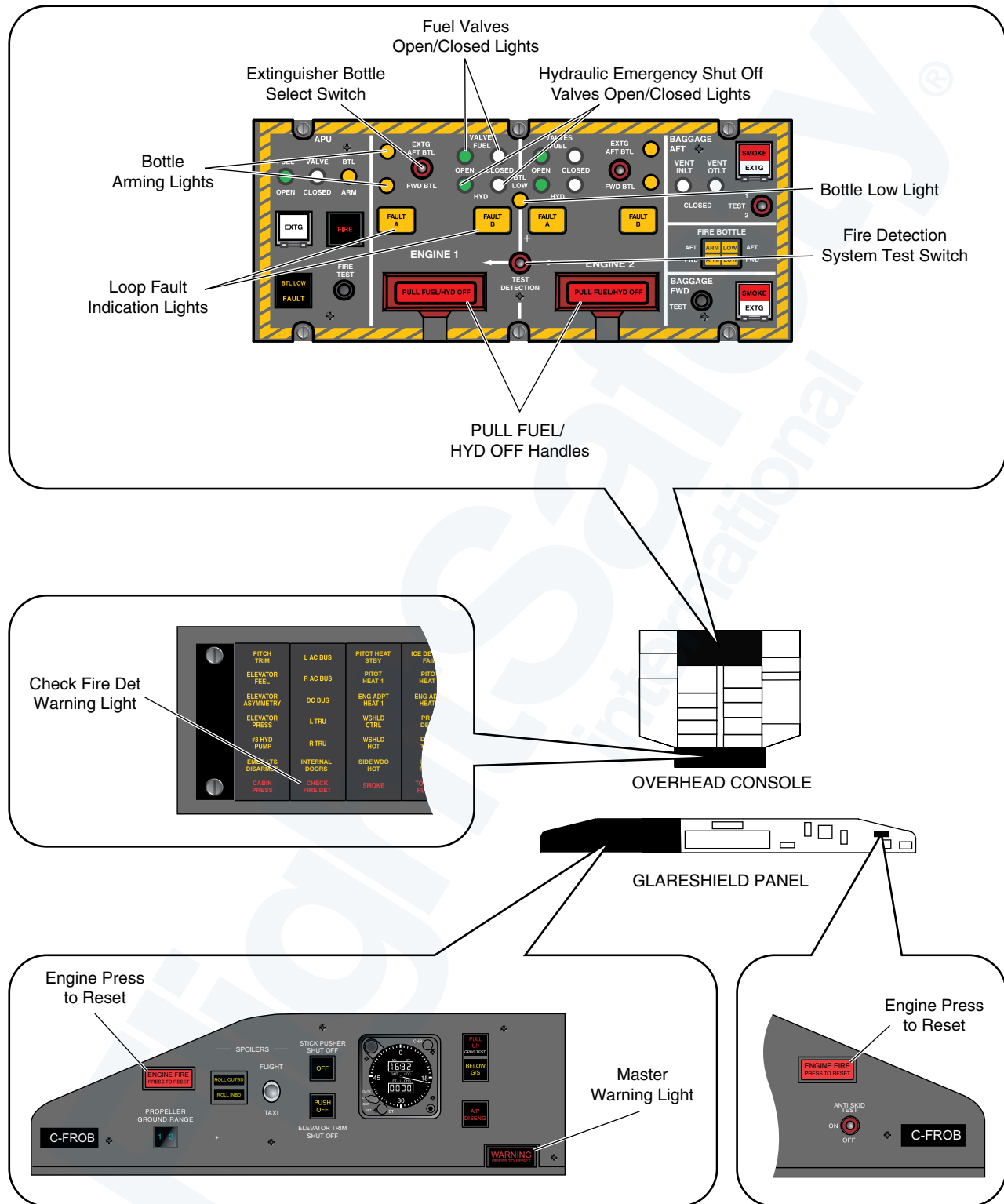
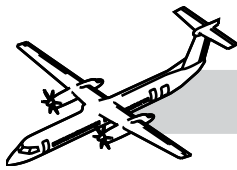
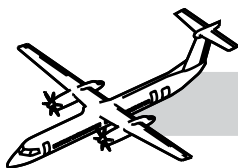


Figure 26-5. Controls and Indications Nacelle Fire Detection



Nacelle Fire Protection Panel (Center)

Refer to:

- Figure 26-5. Controls and Indications Nacelle Fire Detection.
- Figure 26-6. Nacelle Fire Protection Panel (Center).

The center zone of the FPP is divided into left and right sections that correspond to the left and right engine nacelles, respectively. System operation, monitoring and status indications are through the Engine FPP. Switches are provided on the Engine FPP for corrective action for fire condition.

Nacelle FPP indications are provided for:

- Bottle arming (FWD and AFT, Either engine)
- Bottle pressure monitoring (BTL LOW Light)

- Engine alarm indication (Red Lights - PULL FUEL/HYD OFF Handles)
- Fuel and hydraulic valve status (Green - Open, White - Closed)
- Fire detector fault status (Fault A - PEC and/or EPZ, Fault B - MLG).

Nacelle FPP controls are provided for:

- Bottle discharging (FWD and AFT)
- Engine detection and alarm
- Pilot Initiated Test (TEST DETECTION)
- Activation of the fuel and hydraulic valves, "PULL FUEL/HYD OFF" handle.

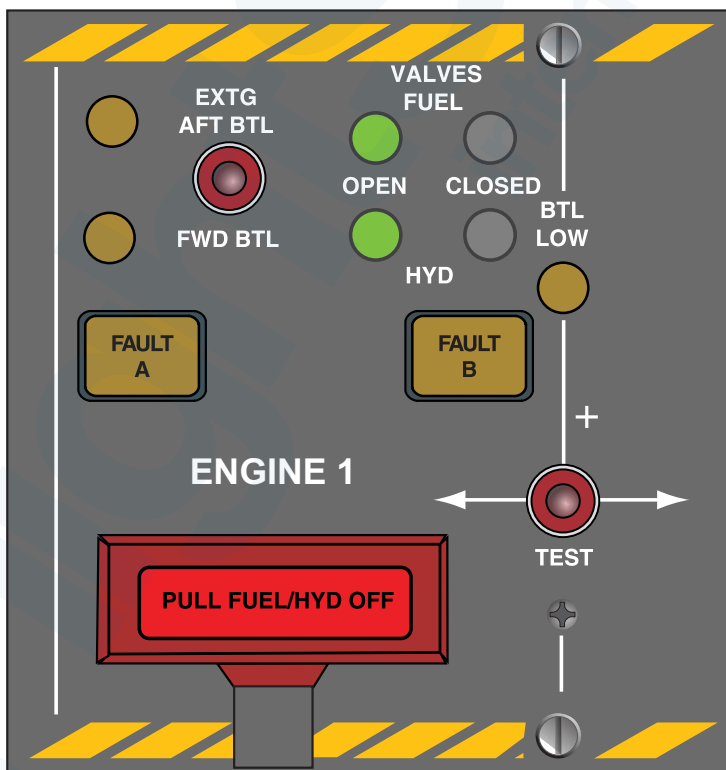
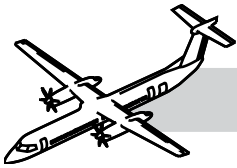


Figure 26-6. Nacelle Fire Protection Panel (Center)



Baggage Compartment Fire Protection Panel (Right)

Refer to Figure 26-7. Baggage Compartment Fire Protection Panel (Right).

The right zone of the FPP is divided into top and bottom sections that correspond to the AFT and FWD baggage compartments, respectively. System operation, monitoring and status indications are through the Baggage Compartment FPP. Switches are provided on the FPP for corrective action for detected smoke conditions.

Baggage Compartment FPP indications are provided for:

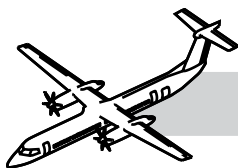
- Extinguisher bottle arming (HRD Bottles)
- Extinguisher bottle pressure monitoring (HRD and LRD)
- Smoke alarm indication (FWD and AFT)
- AFT baggage compartment vent valve status.

Baggage Compartment FPP controls are provided for:

- Activation of extinguisher bottles
- Built-In-Test (BIT).



Figure 26-7. Baggage Compartment Fire Protection Panel (Right)



Auxiliary Power Unit (APU) Fire Protection Panel (Left)

Refer to Figure 26-8. Auxiliary Power Unit Fire Protection Panel.

The left zone of the FPP supplies indications of fire conditions in the APU compartment. APU fire detection and extinguishing is automatic. If the automatic system fails, then a manual override capability is available.

APU FPP indications are provided for:

- Extinguisher bottle arming
- Bottle pressure monitoring
- Fire alarm indication

- Fuel valve status
- Fire detector fault status.

Switches are provided on the FPP for:

- Backup and override extinguishing
- Built-In-Test (BIT).

NOTE

The Engine, APU and Baggage compartment Fire Protection Systems are monitored by the control amplifier, but the control amplifier has been kept out of the critical path for detection and suppression functions.

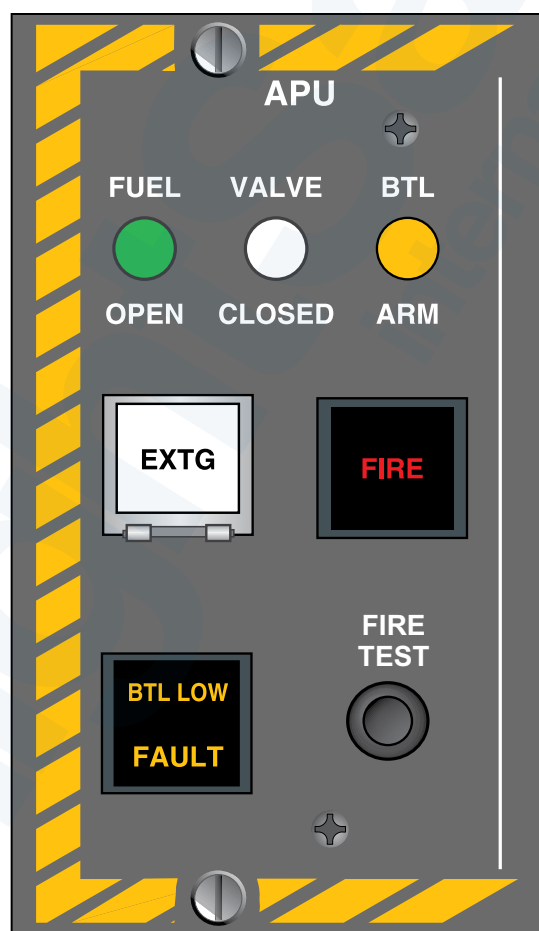
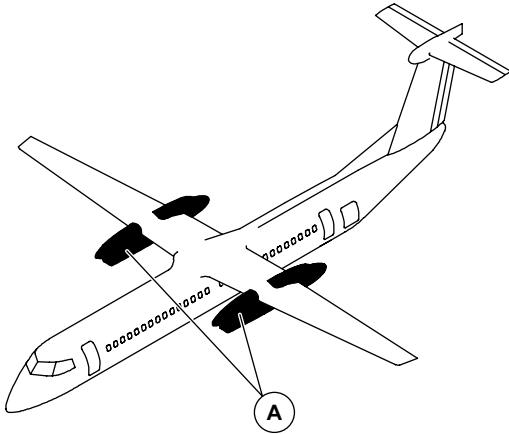
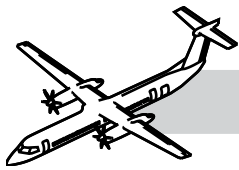
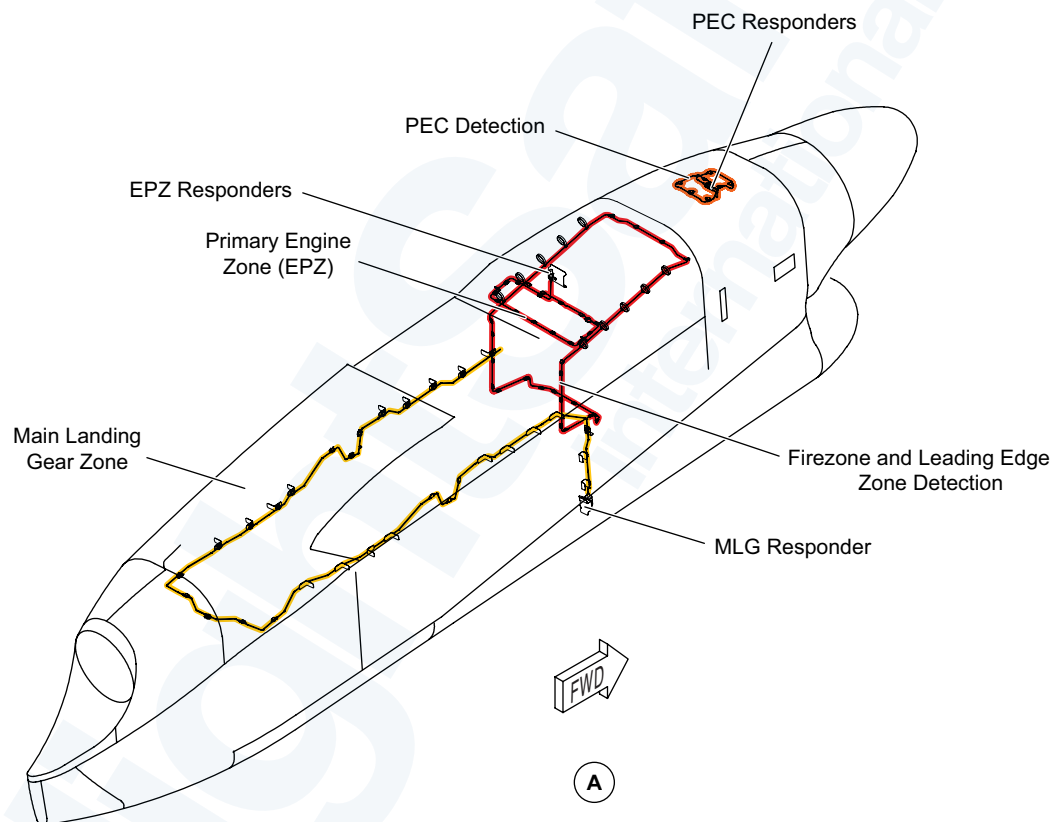


Figure 26-8. Auxiliary Power Unit Fire Protection Panel

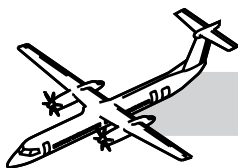
**NOTE**

Left nacelle shown,
right nacelle similar.



█	EPZ Responders
█	PEC Responders
█	MLG Responders

Figure 26-9. Nacelle Fire Detection System



26-11-00 NACELLE FIRE DETECTION SYSTEM

NOTES

INTRODUCTION

The nacelle fire detection system senses fire or overheat conditions in the engine fire zones and supplies indications to the flight compartment.

GENERAL

Refer to Figure 26-9. Nacelle Fire Detection System.

Question: How many APDs in engine nacelle and where are they installed ?

The nacelle fire detection system uses Advanced Pneumatic Detectors (APD) to sense a fire or overheat condition in three engine fire zones in the nacelles. Three APD's are in each nacelle. The APD's provide detection in the Main Landing Gear (MLG) zone, Engine Primary Zone (EPZ) and Propeller Electronic Control (PEC) zone.

The APD's interface with the control amplifier and the FPP in the flight compartment.

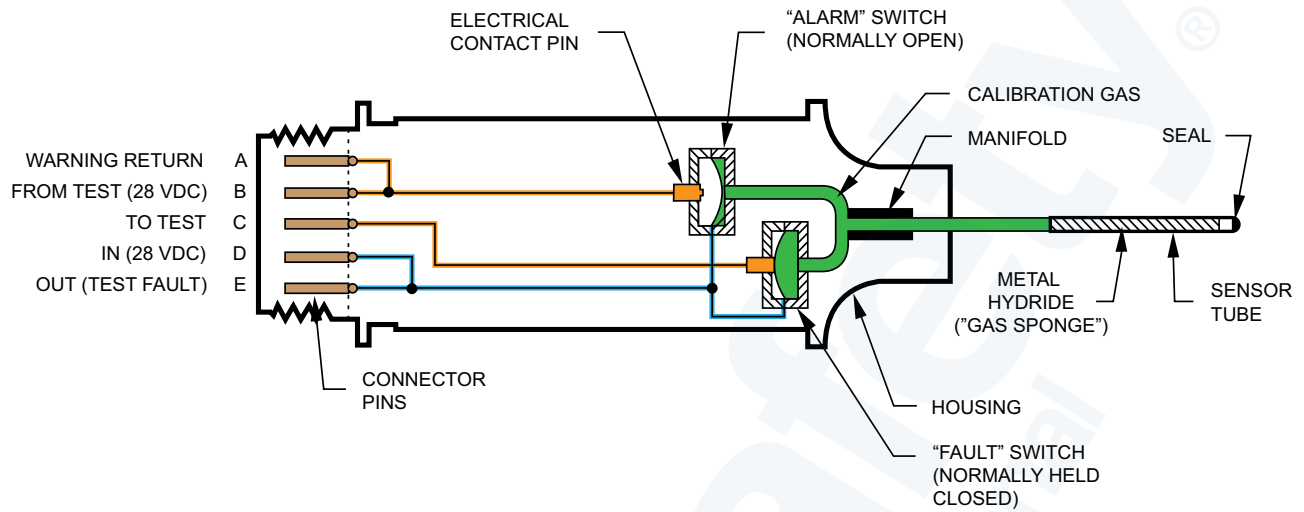
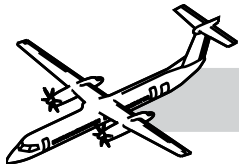


Figure 26-10. Integrity and Alarm - APD

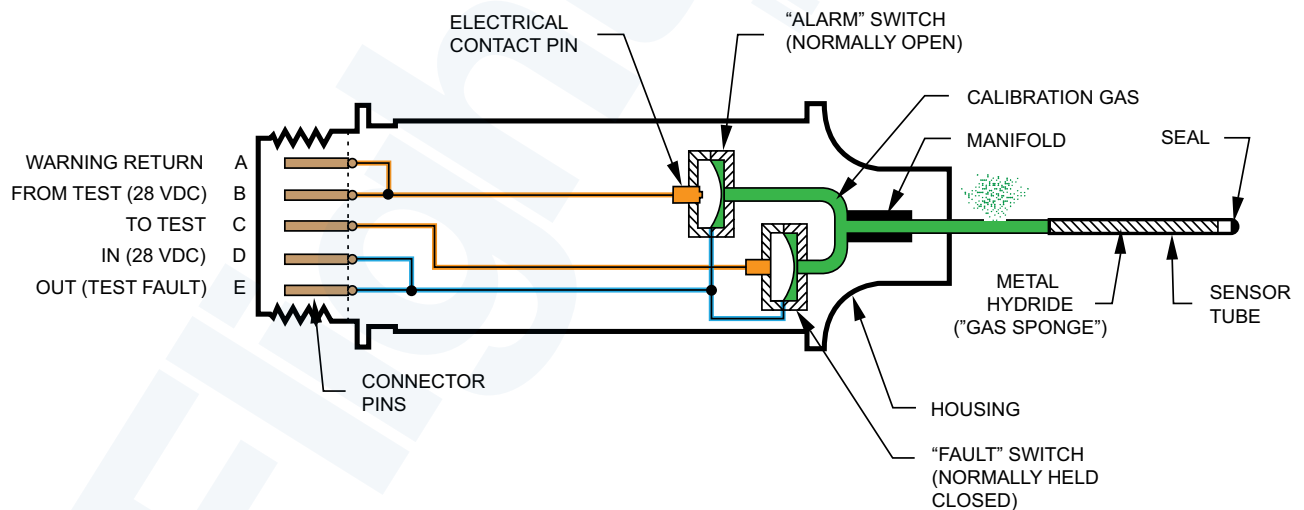
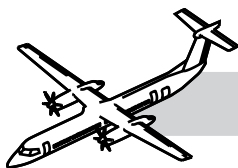


Figure 26-11. Fault A or B - APD



SYSTEM DESCRIPTION

NOTES

The APD has a main element body and a sensor tube which extends around the engine fire zone. The responder body is held in place by two P-clamps; the sensor tube is supported by additional clamps and grommets. The sensor tube is filled with helium (inert gas) which is sensitive to temperature changes in the respective engine fire zones.

Refer to Figure 26-10. Integrity and Alarm - APD.

Two switches (integrity and alarm) are integral in the APD.

Refer to Figure 26-11. Fault A or B - APD.

The integrity switch, normally closed by the helium pressure, opens if this pressure is lost. This signals the control amplifier to turn on the corresponding FAULT A or FAULT B lights on the FPP. The CHK FIRE DET light illuminates concurrently with a FAULT light indication.

The alarm switch is normally open, and closes when the pressure increases to a factory set level based on the element length and temperature in each zone.

NOTE

There are two ways in which the APD can produce an alarm condition:

- Whole length heating or averaging
- Short length heating or discrete.

Whole Length Heating or Averaging

In this condition, the helium gas in the sealed sensor tube expands. Since the sensor tube is a constant volume enclosure, the increased pressure is sensed by the alarm switch. When the pressure increases to the factory set level, it operates the alarm switch. When the alarm switch closes, the fire indication lights on the FPP will come on.

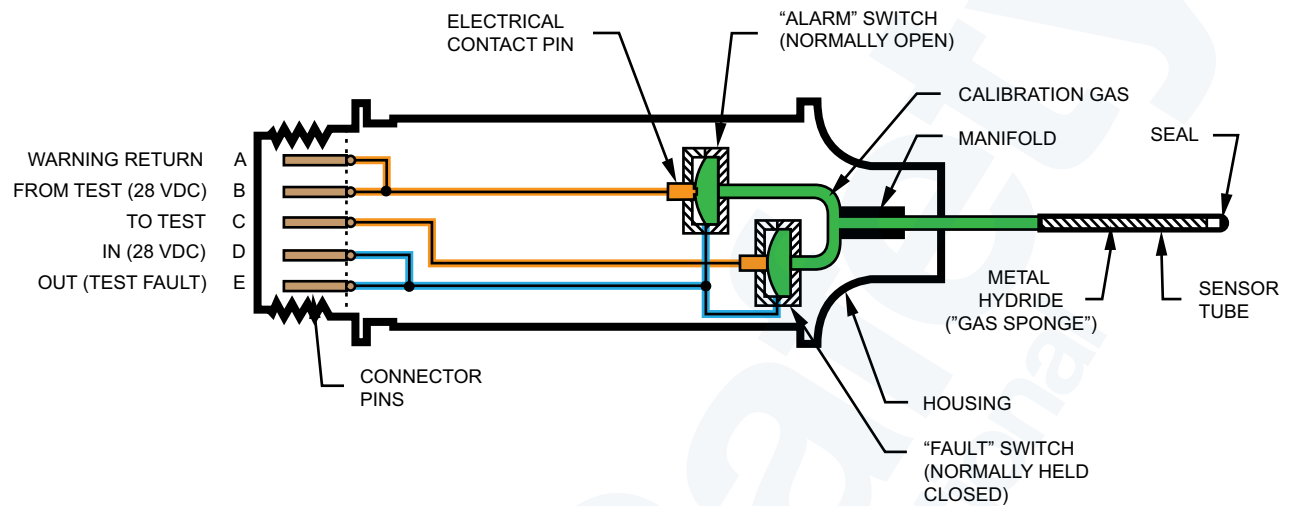
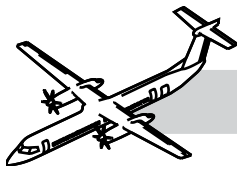


Figure 26-12. Short Length Heating or Discrete - APD

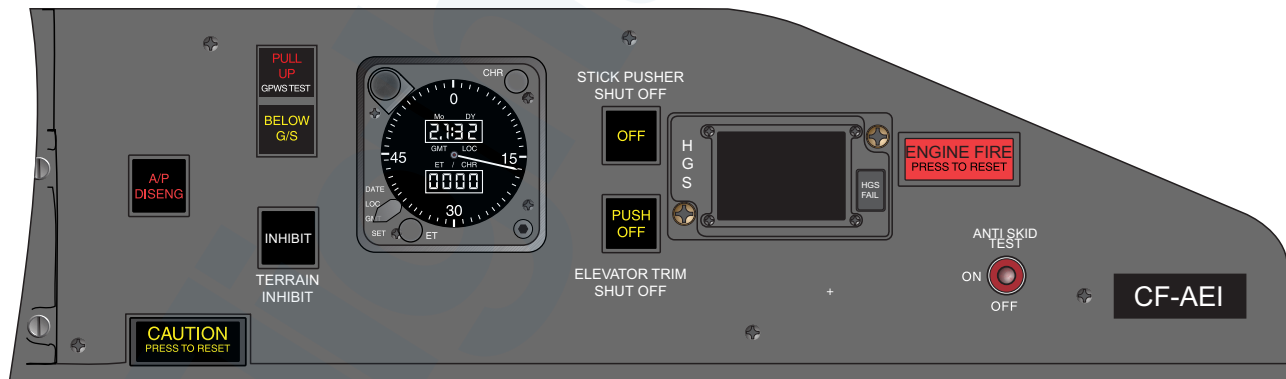
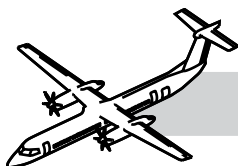


Figure 26-13. Engine Fire Press to Reset



Short Length Heating or Discrete

NOTES

Refer to Figure 26-12. Short Length Heating or Discrete - APD.

In this condition, the metallic titanium core releases the absorbed hydrogen gas. The gas is released above a critical temperature. The pressure in the sensor tube increases quickly above this critical temperature, operating the alarm switch. When the alarm switch closes, the fire indication lights on the FPP will come on.

Refer to Figure 26-13. Engine Fire Press to Reset.

The APD will generate a fire signal to the appropriate engine PULL FUEL/HYD OFF handle light directly. This enables the fire detection system to give fire warning even if the control amplifier fails or becomes faulty during flight operations.

Simultaneous fire indications will be provided by the control amplifier to signal the:

- CHK FIRE DET
- Master WARNING (flashing)
- ENGINE FIRE PRESS TO RESET indicator
- Warning lights and
- An optional audible bell.

Pushing either the left or right ENGINE FIRE PRESS TO RESET indicator turns the audible bell warning off; Both ENGINE FIRE PRESS TO RESET indicators remain on steady state for the duration of the alarm condition.

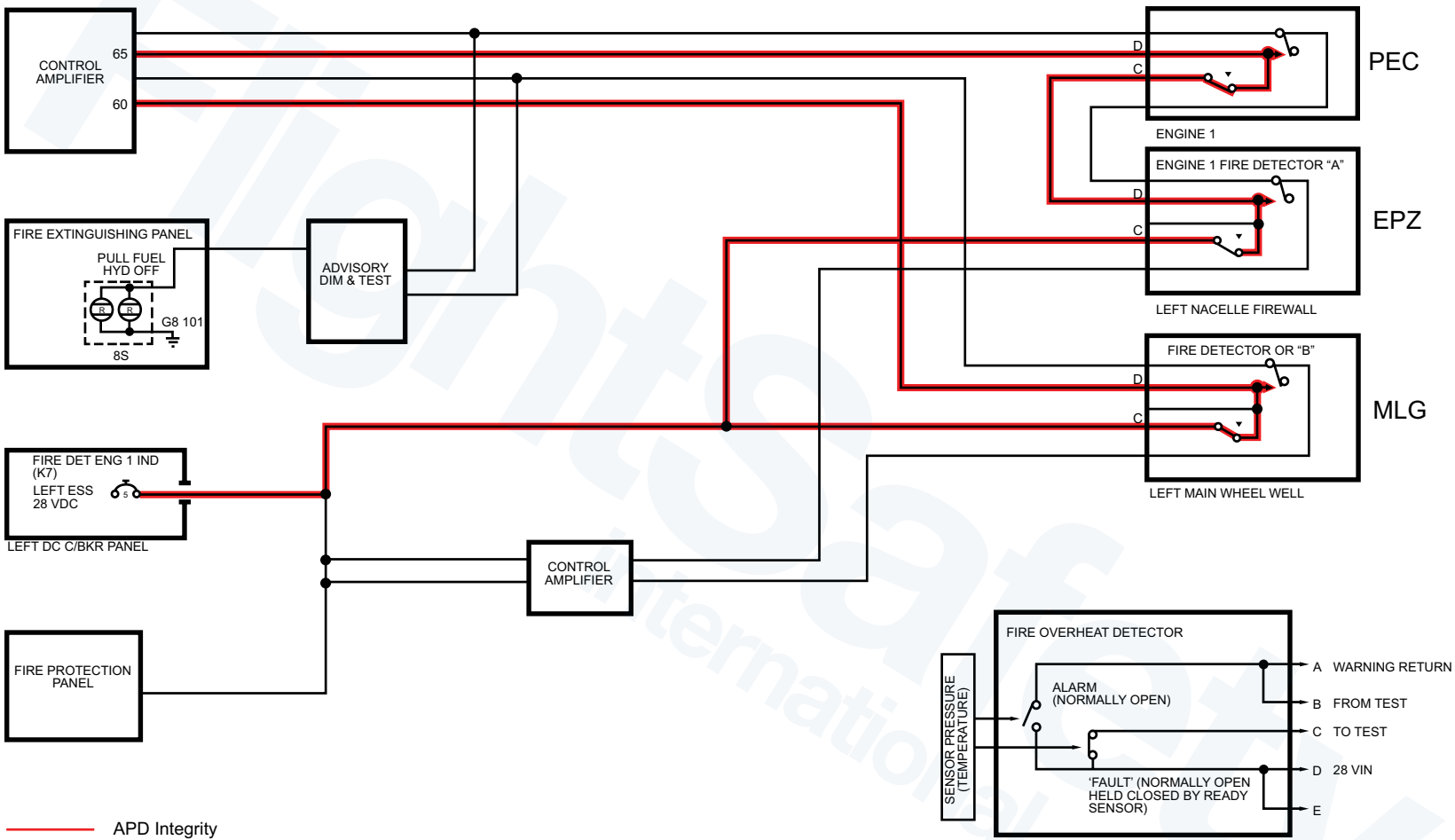
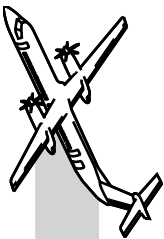
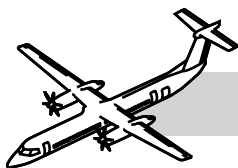


Figure 26-14. Fire Detection - Normal Operation



OPERATION

NOTES

Normal System Operation

Detect

Refer to Figure 26-14. Fire Detection - Normal Operation.

NOTE

Engine number 1 shown, engine number 2 operations similar.

28 VDC is supplied from the left essential bus via circuit breaker K7 (FIRE DET ENG 1 IND) through disconnects and configuration connectors (removed for clarity) to the APD pin C.

This signal passes through the fault switch that is normally held closed by pressure within the APD and exits on pin D. The voltage from the main wheel well detector is then applied to the fire control amplifier, pin 60. EPZ & PEC integrity switches are connected in series. The voltage that exits pin D on the EPZ is applied to the PEC pin C before being sent to the control amplifier pin 65.

The voltage sensed on pins 60 and 65 confirms the integrity of the nacelle fire detection control loops (APD).

If an APD should lose its pressure or there is a wiring fault, the integrity voltage would not be sensed at the control amplifier. A loss of voltage on pin 60 causes the control amplifier to illuminate the FAULT "B" advisory light on the FPP. A loss of voltage on pin 65 causes the control amplifier to illuminate the FAULT "A" advisory light.

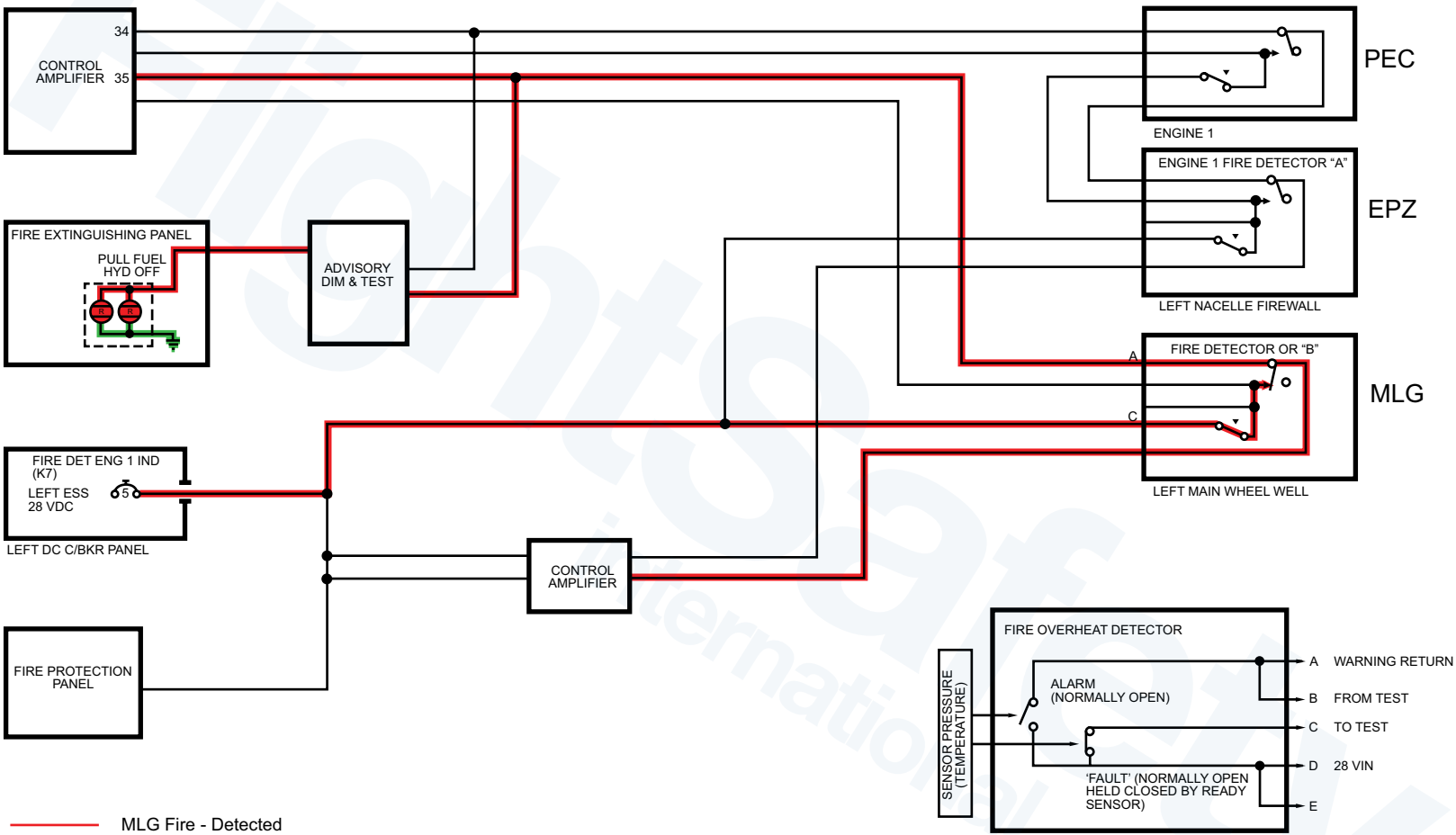
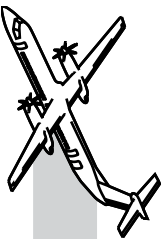
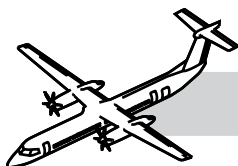


Figure 26-15. Fire Overheat Detection - MLG Fire Detected



Fire/Overheat Detection

NOTES

Refer to Figure 26-15. Fire Overheat Detection
- MLG Fire Detected.

NOTE

MLG shown, EPZ/PEC
operation similar.

At a preset temperature the increasing pressure within the APD reaches a point where the alarm switch closes. 28 VDC is now sent through pin A of the APD. This voltage is sent directly to the PULL FUEL/HYD OFF handle on the FPP illuminating the lights (RED) within the handle. The voltage is also sent to the control amplifier pins 34 or 35 to initiate the other associated warnings.

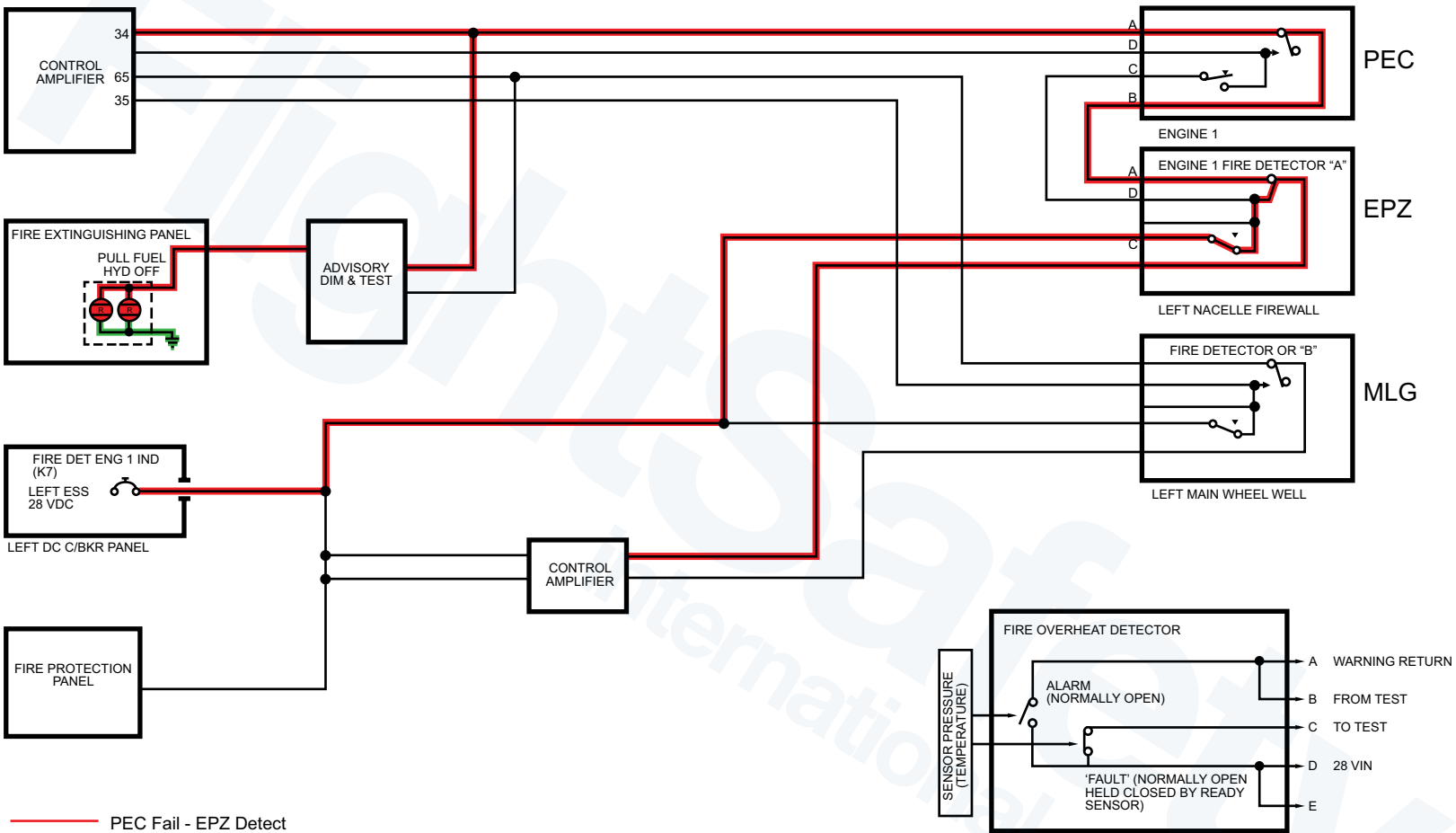
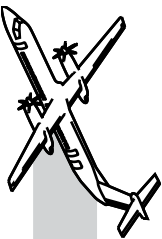
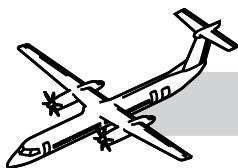


Figure 26-16. PEC Fail -EPZ Detected



Fault and Detect

NOTES

Refer to:

- Figure 26-16. PEC Fail -EPZ Detected.
- Figure 26-17. EPZ Fail - PEC Fail.

The EPZ and PEC APD's are electrically connected in such a way that it is possible to have a fault on the system and still detect a fire/overheat condition.

If PEC APD developed a leak or there was a wiring fault with the integrity circuit, the 28 VDC would not be sensed on pin 65 of the control amplifier.

The control amplifier would illuminate the Fault "A" advisory light on the FPP. If a fire/overheat condition was then sensed by EPZ, the alarm switch would close. Voltage would be sent from EPZ responder pin A to PEC responder pin B. The voltage would then be sent from PEC responder pin A directly to the PULL FUEL/HYD OFF handle on the FPP illuminating the lights (RED) within the handle. The voltage is also sent to the control amplifier pins 34 to initiate the other associated warnings.

If EPZ APD developed a leak or there was a wiring fault with the integrity circuit, the 28 VDC would not be sensed on pin 65 of the control amplifier. The control amplifier would illuminate the Fault "A" advisory light on the FPP.

If PEC then detected a fire/overheat condition, the switch closes but there is no voltage available to signal the control amplifier.

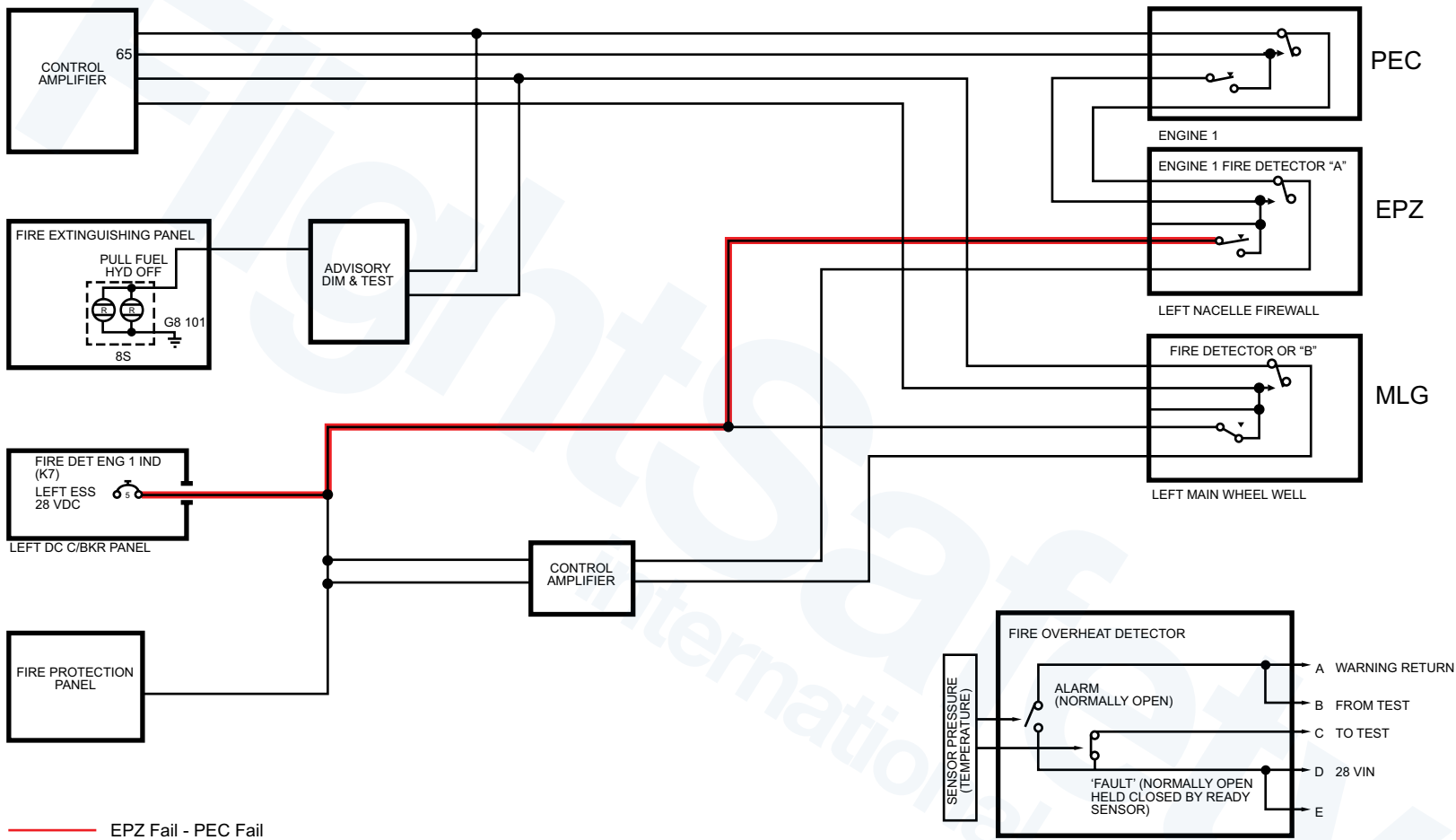
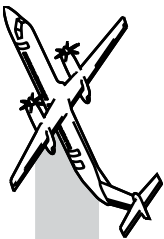
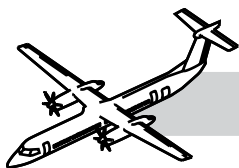


Figure 26-17. EPZ Fail - PEC Fail



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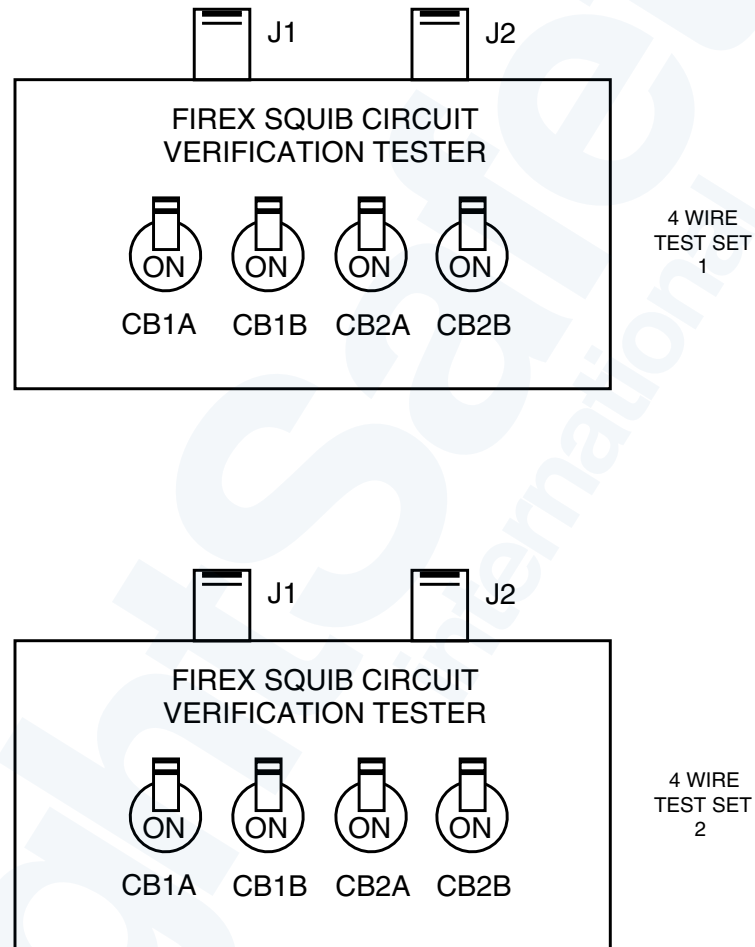
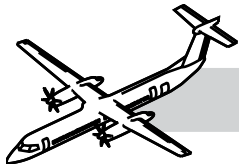
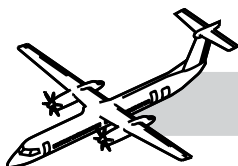


Figure 26-18. 4 Wire Firex Squib Circuit Verification Testers



FUNCTIONAL TEST OF THE NACELLE FIRE EXTINGUISHING SYSTEM

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Refer to Figure 26-18. 4 Wire Firex Squib Circuit Verification Testers.

CAUTION

OBEY ALL ELECTROSTATIC DISCHARGE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR DEVICES SENSITIVE TO ELECTROSTATIC DISCHARGE. IF YOU DO NOT DO THIS YOU CAN CAUSE DAMAGE TO EQUIPMENT.

WARNING

THE DISCHARGE CARTRIDGE (SQUIB) IS AN ELECTRICALLY FIRED EXPLOSIVE DEVICE.

WHEN YOU DO MAINTENANCE MAKE SURE THAT THE PINS OF THE ELECTRICAL CONNECTOR (SQUIB) HAVE THE SAME ELECTRICAL POTENTIAL.

INSTALL A SHUNT PLUG ON THE ELECTRICAL CONNECTOR OF THE SQUIB IMMEDIATELY AFTER YOU DISCONNECT THE ELECTRICAL CONNECTION.

IF YOU DO NOT DO THIS YOU CAN CAUSE INJURIES TO PERSONNEL IF THE

SQUIB IS ACCIDENTLY ENERGIZED.

Connect test set 1 to the No.1 nacelle fire extinguisher bottle.

Connect test set 2 to the No.2 nacelle fire extinguisher bottle.

Remove the electrical connectors of the forward and aft extinguisher bottles, and identify the connectors by attaching labels.

Install shunt plugs on the electrical connectors of the cartridges (squibs).

Open the circuit breakers as listed in the TASK.

Do a functional test on the nacelle fire extinguishing system by pulling the PULL FUEL/HYD HANDLES and observing the correct indications on the test set and the FPP.

Remove the test set.

Remove the shunts.

Re-install the labeled connectors.

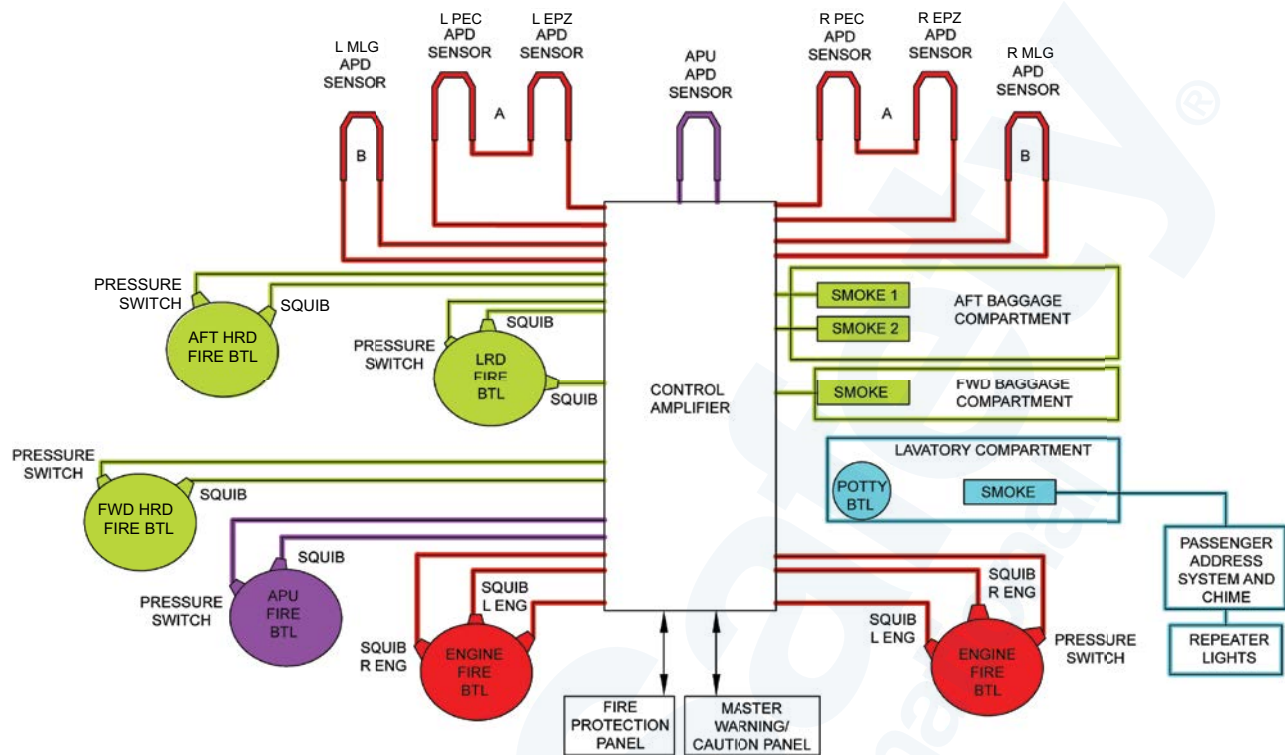
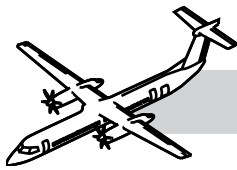
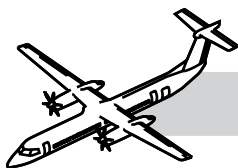


Figure 26-19. Fire Protection System Schematic - Extinguishing



Figure 26-20. Nacelle Fire Extinguishing Bottles



Nacelle Fire Extinguisher Bottles

Refer to:

- Figure 26-19. Fire Protection System Schematic - Extinguishing.
- Figure 26-20. Nacelle Fire Extinguishing Bottles.
- Figure 26-21. Nacelle Fire Extinguisher Bottles Schematic.

There are two dual port fire extinguishing bottles installed in a FWD and AFT configuration in the left wing root for engine fire suppression.

Each bottle is connected through a Check Tee with a drain to the distribution tubes. This gives extinguisher coverage to the right or left nacelle. Electrical connections are installed on the two discharge valves with the explosive squibs and the bottle monitor pressure switch.

The fire bottles are stainless steel spheres charged with Halon 1301 as an agent and pressurized with Nitrogen to 600 to 625 psig at 70° F (21°C).

Each bottle has two discharge valves actuated by cartridges (squibs) in a housing assembly with an electrical connection.

The bottles use dual-bridgewire Electro-Explosive Devices (EED) within each cartridge and redundant power lines with separate circuit breakers for reliability.

A pressure switch is installed to provide indication for low bottle pressure, set to open at 315 psig maximum (increasing) and to close at 225 ± 25 psig (decreasing).

One discharge valve on each bottle is connected to the distribution system of Nacelle No.1, and the other is connected to the distribution system of Nacelle No.2.

Thermal relief is provided through one of the two burst discs. The relief is then routed through the distribution system into the respective nacelles.

NOTE

Use of Halon 1301 does not require maintenance clean up after a discharge into the nacelle.

A manual exercise key access in the pressure switch is supplied for ground check of proper pressure switch operation. The key will open the pressure switch contacts, to simulate a depressurized bottle.

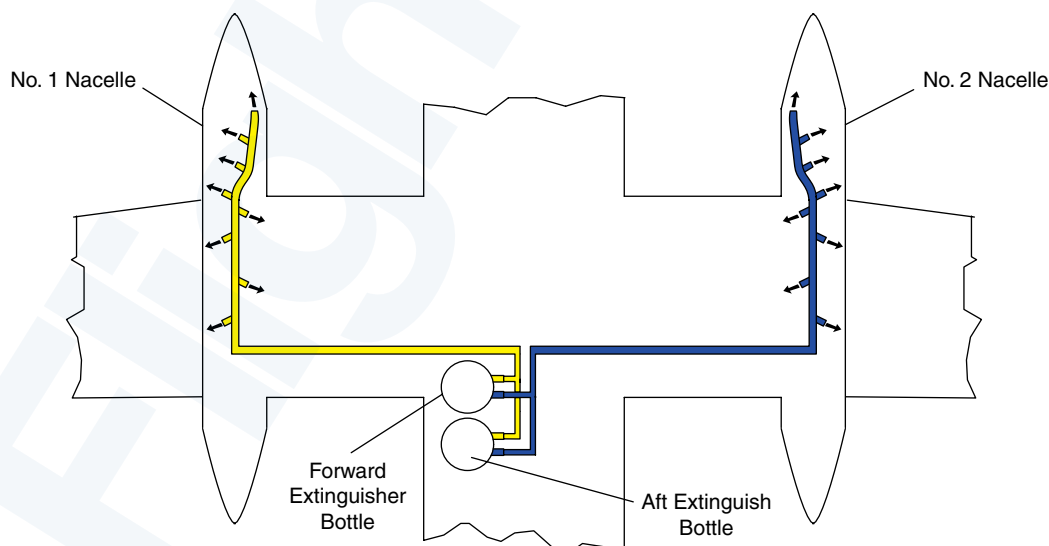
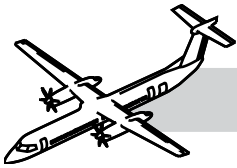
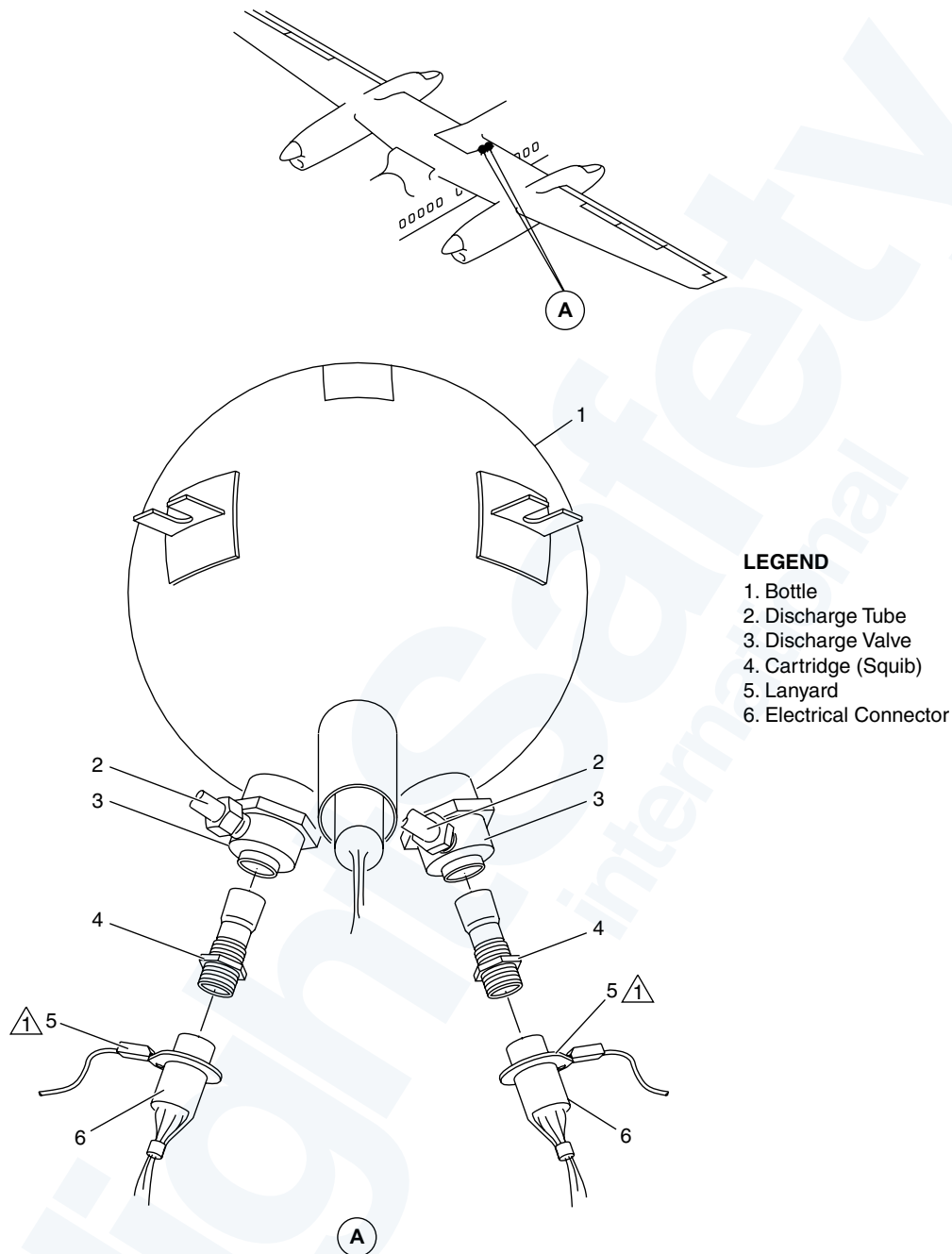
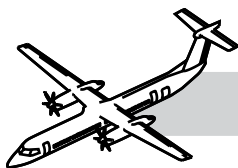


Figure 26-21. Nacelle Fire Extinguisher Bottles Schematic


DASH 8 Q400 MAINTENANCE TRAINING MANUAL

Figure 26-22. Nacelle Fire Extinguisher Bottle and Cartridges



Fire Extinguisher Cartridges

NOTES

Refer to:

- Figure 26-22. Nacelle Fire Extinguisher Bottle and Cartridges.
- Figure 26-23. LH Underwing Fire Bottle Pipes.

The fire extinguisher cartridges are installed on the fire extinguisher bottles.

NOTE

Extreme care has to be taken when handling the fire extinguishing bottles on the aircraft during maintenance activity. Refer to the appropriate procedures in the Aircraft Maintenance Manual Chapter 26-21-01 and 26-21-06 Fire Protection.

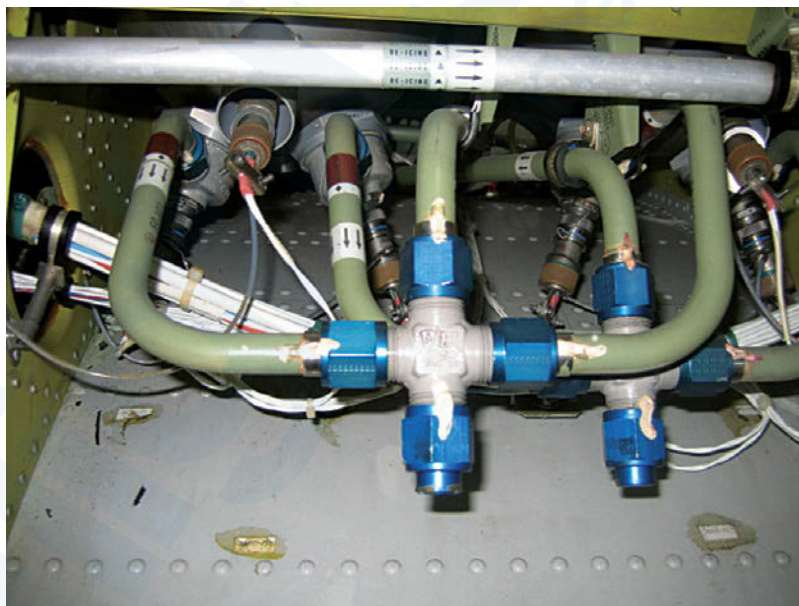


Figure 26-23. LH Underwing Fire Bottle Pipes

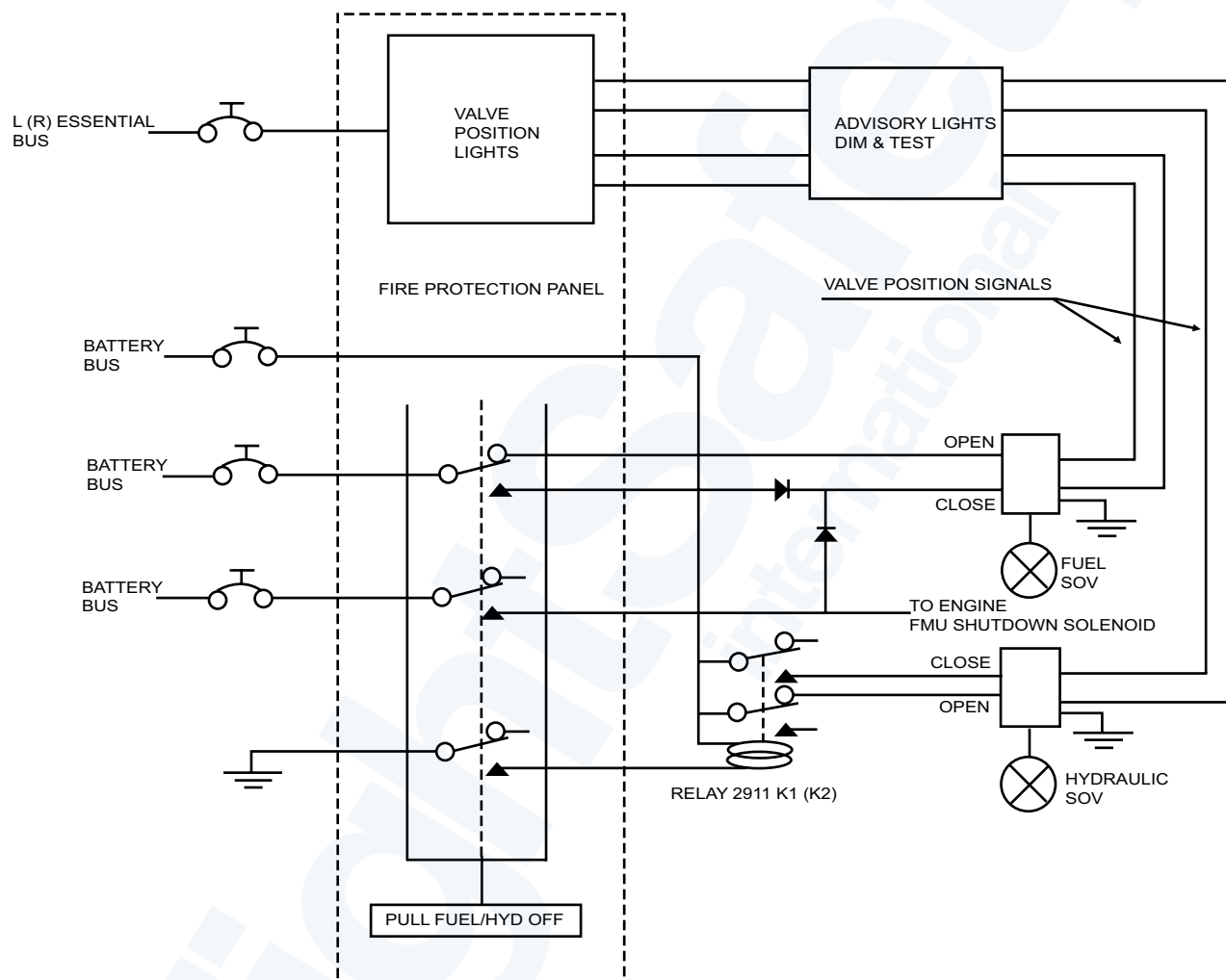
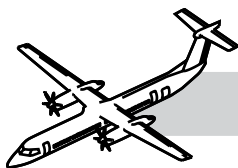


Figure 26-24. Flammable Fluid Shut-off System - Electrical Schematic



Fire Extinguishing

NOTES

Refer to Figure 26-24. Flammable Fluid Shut-off System - Electrical Schematic.

PULL FUEL/HYD OFF

The initial step for engine fire extinguishing event starts with the flammable fluid shut-off system.

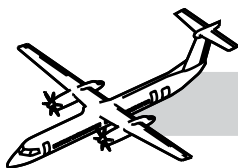
For each engine the system comprises the following:

- The PULL FUEL/HYD OFF T-handle on the FPP
- Fuel and hydraulic valve position status lights on the FPP
- A motorized fuel SOV on the wing rear spar outboard of the rib at station Y112
- A motorized hydraulic SOV on the right wall of the nacelle aft of the firewall (Zone 6).

Fuel SOV - For details on the valve refer to AMM Ch 28-21-00.

Hydraulic SOV - For details on the valve refer to AMM Ch 29-11-06.





SOV Position Indication

Each SOV has position switches for the closed and open positions. These switches drive lights on the FPP via the Advisory Lights Dim and Test unit.

White light indicates valve CLOSE, and green light indicates valve OPEN. For the fuel SOV only, the position switch wiring is duplicated and separated in the fuselage to meet the rotor burst requirements.

Nacelle Fire Extinguishing

Refer to Figure 26-25. Nacelle Fire Extinguishing System - Electrical Schematic (Sheet 1 of 2).

NOTE

Engine number 1 shown, engine number 2 operations similar.

When the PULL FUEL/HYD OFF handle is pulled, five internal contacts move from their normally closed position to the normally open position. These contacts perform the following:

- 1C - Provides a ground through pin S to hydraulic system No.1. Close the hydraulic SOV.
- 2C - Supplies 28 VDC from the battery bus to close the engine No.1 fuel SOV.
- 3C - Supplies 28 VDC from the battery bus to close the engine No.1 fuel shut-off valve. Supplies 28 VDC from the battery bus to close the airframe shut down solenoid in the No.1 engine Fuel Metering Unit (FMU).
- 4C – Supplies 28 VDC from the battery bus to the control amplifier – Signal for Bottle Arm. Supplies 28 VDC from the battery bus to the Firex switch.
- 5C – Supplies 28 VDC from the battery bus to the control amplifier – Signal for Bottle Arm. Supplies 28 VDC from the battery bus to the Firex switch.

NOTE

There is a dual input of 28 VDC to the Firex switch for redundancy.

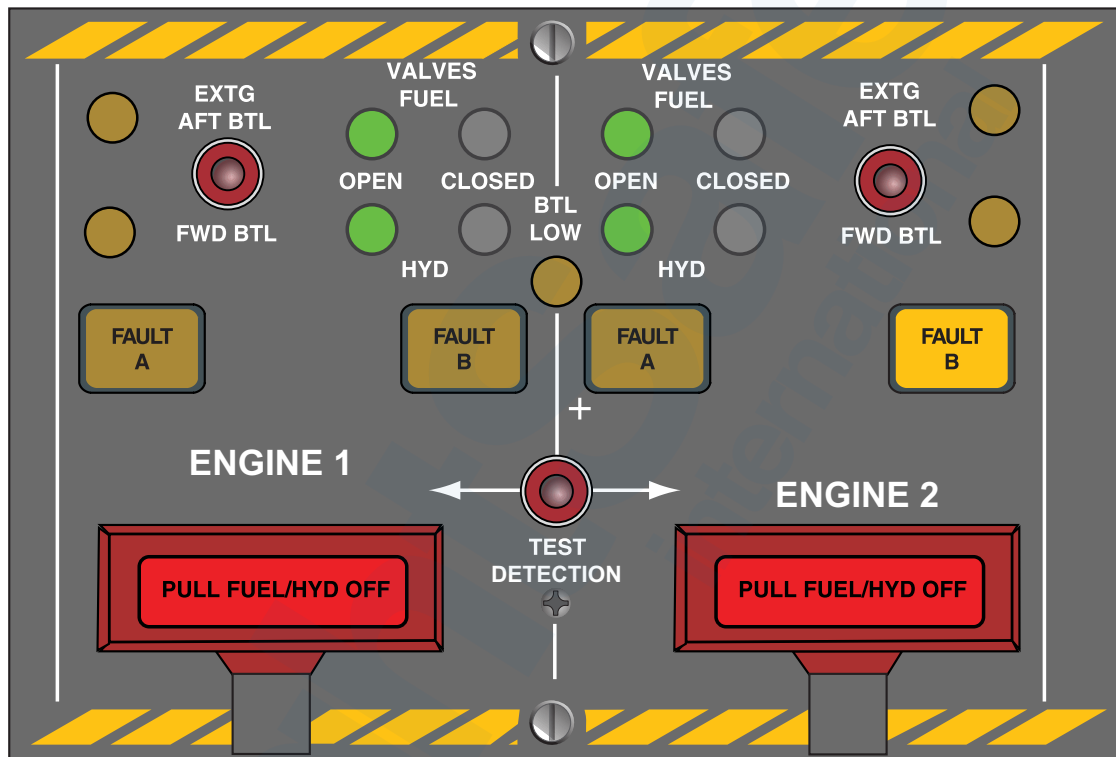
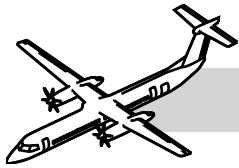
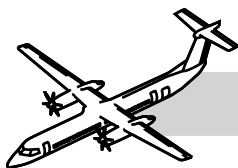


Figure 26-26. Fire Protection Panel - Nacelle



Refer to Figure 26-26. Fire Protection Panel - Nacelle.

Operation of the system is from the FPP.

The bottle to fire is selected by operating the EXTG switch towards the AFT BTL or FWD BTL position. The selected bottle will discharge its contents into all 3 zones (EPZ, PEC & MLG) simultaneously. The bottle pressure switch will open and the engine "BTL LOW" indicator light on the FPP will come on, thus indicating loss of bottle pressure in one or both bottles.

OPERATIONAL TEST OF NACELLE FIRE DETECTION SYSTEM

Refer to the Bombardier AMM PSM 1-84-2 for a detailed description of this maintenance practice.

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Ensure the CB's listed in the task sheet are closed.

Make sure the aircraft electrical system is energized.

Do a Caution/Advisory lights test to check that all the lights are working.

On the FPP hold the TEST DETECTION switch to ENGINE No.1. Check as follows:

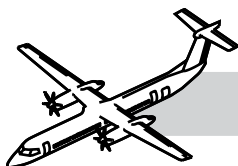
- The CHECK FIRE DET red warning light flashes
- The master WARNING LIGHT flashes
- Both the pilot and co-pilot ENGINE FIRE PRESS TO RESET lights flash
- The ENGINE No.1 PULL FUEL/HYD OFF handle red light comes ON
- The ENGINE No.1 FAULT A amber light comes ON
- The ENGINE No.1 FAULT B amber light comes ON
- Make sure an intermittent audible warning tone is heard
- Release TEST DETECTION switch to the center position.

Pull the ENGINE No.1 PULL FUEL/HYD OFF HANDLE. Check as follows:

- ENG 1 FWD BTL arm amber light is ON
- ENG 1 AFT BTL arm amber light is ON
- FUEL VALVE CLOSED white light is ON
- HYD VALVE CLOSED white light is ON
- Push the ENGINE No.1 PULL FUEL/HYD OFF HANDLE back in.

Repeat process for Engine No.2.





Refer to Figure 26-27. Nacelle Fire Extinguishing System - Electrical Schematic (Sheet 2 of 2).

NOTES**NOTE**

Aft Bottle Squib shown Fwd
Bottle similar.

When the Firex switch is selected to AFT BTL position 28 VDC from the Batt Bus through the Fire Ext Cont Amp 1 & 2 circuit breaker is applied to pin 3 of the Aft Firex Squib. 28 VDC from the Batt Bus through Fire Ext Cont Cargo circuit breaker is applied to pin 1 of the Aft Firex Squib. This redundancy is to ensure the squib will fire releasing the extinguishing agent.

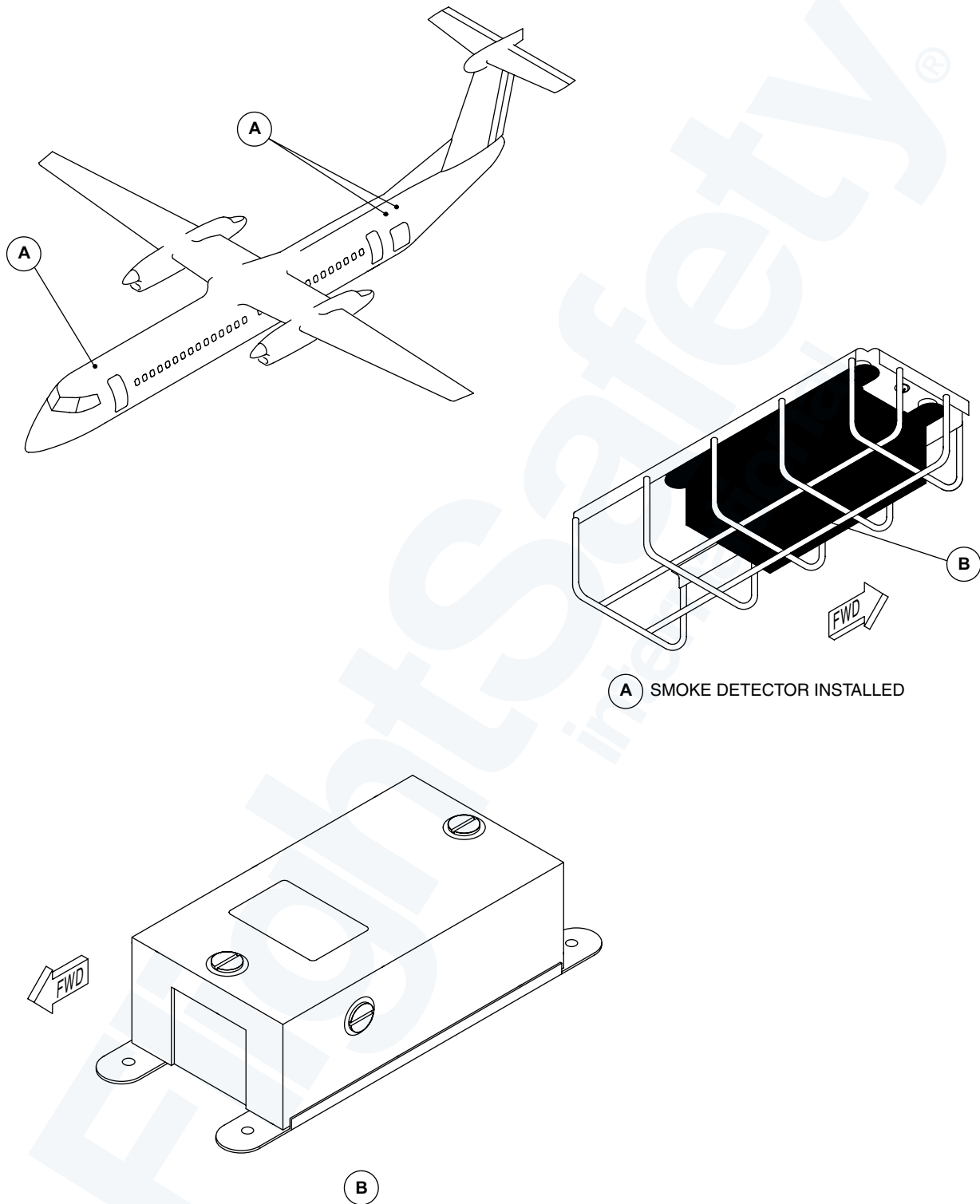
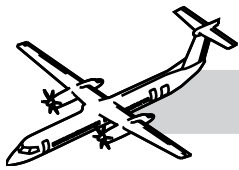
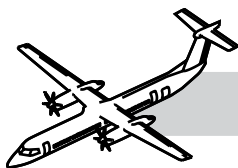


Figure 26-28. Smoke Detector - Locations



26-12-00 SMOKE DETECTION SYSTEM

INTRODUCTION

Smoke detection is done by smoke detectors which are installed in the AFT baggage (cargo) compartment, the FWD baggage area and the lavatory.

The smoke detectors in the FWD and AFT baggage compartments are attached to the mounting surface with four pan-head screws. Case grounding for the smoke detectors are provided through the bottom area of the base plate.

GENERAL

Refer to:

- Figure 26-28. Smoke Detector - Locations.
- Figure 26-29. Aft Cargo Fwd Smoke Detector.
- Figure 26-30. Lavatory Smoke Detector.

If a smoke condition is sensed, the applicable warning indicator lights on the FPP or the repeater lights in the passenger compartment ceiling come on.

The smoke detection system has the components that follow:

- AFT baggage smoke detectors
- FWD baggage smoke detector
- Lavatory smoke detector.



Figure 26-29. Aft Cargo Fwd Smoke Detector



Figure 26-30. Lavatory Smoke Detector

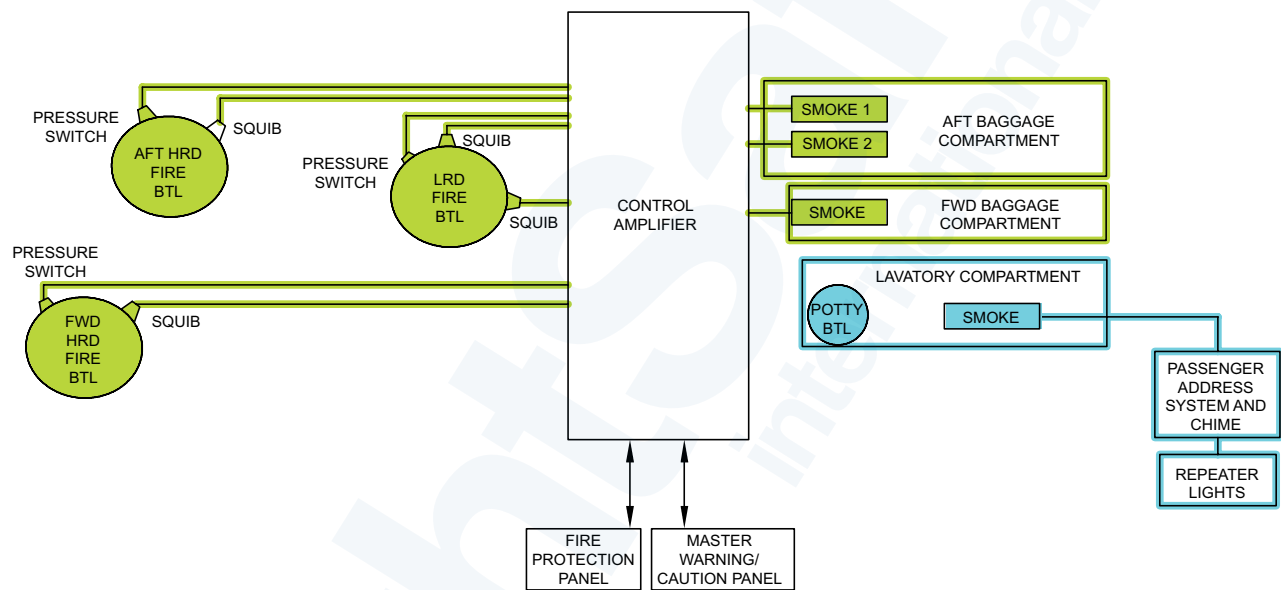
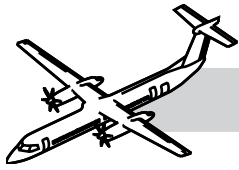
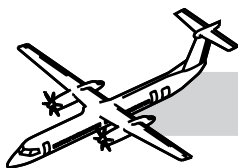


Figure 26-31. Fire Protection System Schematic - Extinguishing



SYSTEM DESCRIPTION

NOTES

Refer to Figure 26-31. Fire Protection System Schematic - Extinguishing.

The AFT and the FWD baggage smoke detectors are monitored by the control amplifier. If a smoke condition is sensed, each baggage smoke detector directly turns on the applicable AFT or FWD SMOKE (red) FPP advisory light. The alarm signal is also sent to the control amplifier. The control amplifier illuminates the SMOKE warning light, master WARNING PRESS TO RESET switch and applicable FPP AFT or FWD BOTTLE ARM (amber) light.

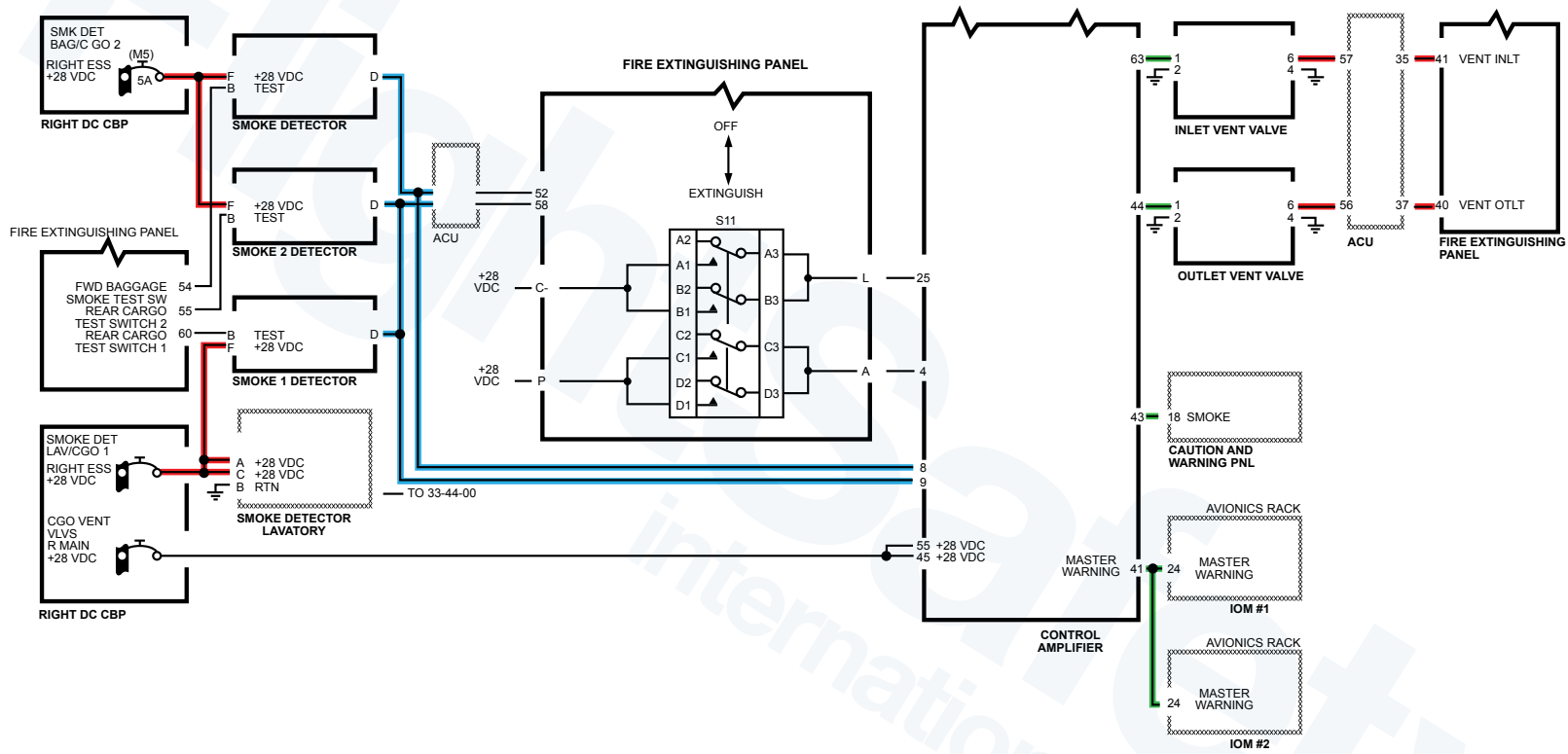
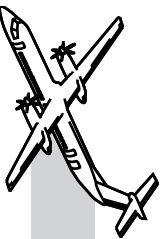
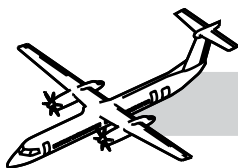


Figure 26-32. Smoke Detected



Detection

Smoke Detected

Refer to Figure 26-32. Smoke Detected.

The FWD baggage smoke detector and the AFT baggage smoke detector 2 (Aft) receive 28 VDC from the right essential bus via circuit breaker M5, SMK DET BAG/CGO 2. The AFT baggage smoke detector 1 (Fwd) and the LAV smoke detector receive 28 VDC from the right essential bus via circuit breaker L5, SMK DET LAV/CGO.

Smoke detected by the FWD baggage smoke detector outputs 28 VDC on pin D directly to the FPP to illuminate the SMOKE (Red) advisory switchlight. This signal is also sent to the fire control amplifier pin 8. The control amplifier outputs to illuminate the following:

- The SMOKE red switchlight
- The EXT white switchlight
- The FWD ARM amber light and
- The SMOKE warning light & Master WARNING.

Smoke detected by either AFT baggage smoke detector outputs 28 VDC on pin D directly to the FPP to illuminate the SMOKE (Red) advisory switchlight.

This signal is also sent to the fire control amplifier pin 9. The control amplifier outputs to close the inlet and the outlet valve and to illuminate the following:

- The SMOKE red switchlight
- The EXT white switchlight
- The AFT ARM amber light
- The SMOKE warning light & Master WARNING.

The smoke detection system will automatically arm the appropriate squibs if the squibs have bridgewire continuity and power. The smoke detection system will reset upon removal of the smoke. However, the inlet and outlet valves in the AFT baggage compartment will remain closed until the appropriate circuit breakers are reset.

Refer to Figure 26-33. Fire Protection Panel Aft Baggage Smoke Detection.

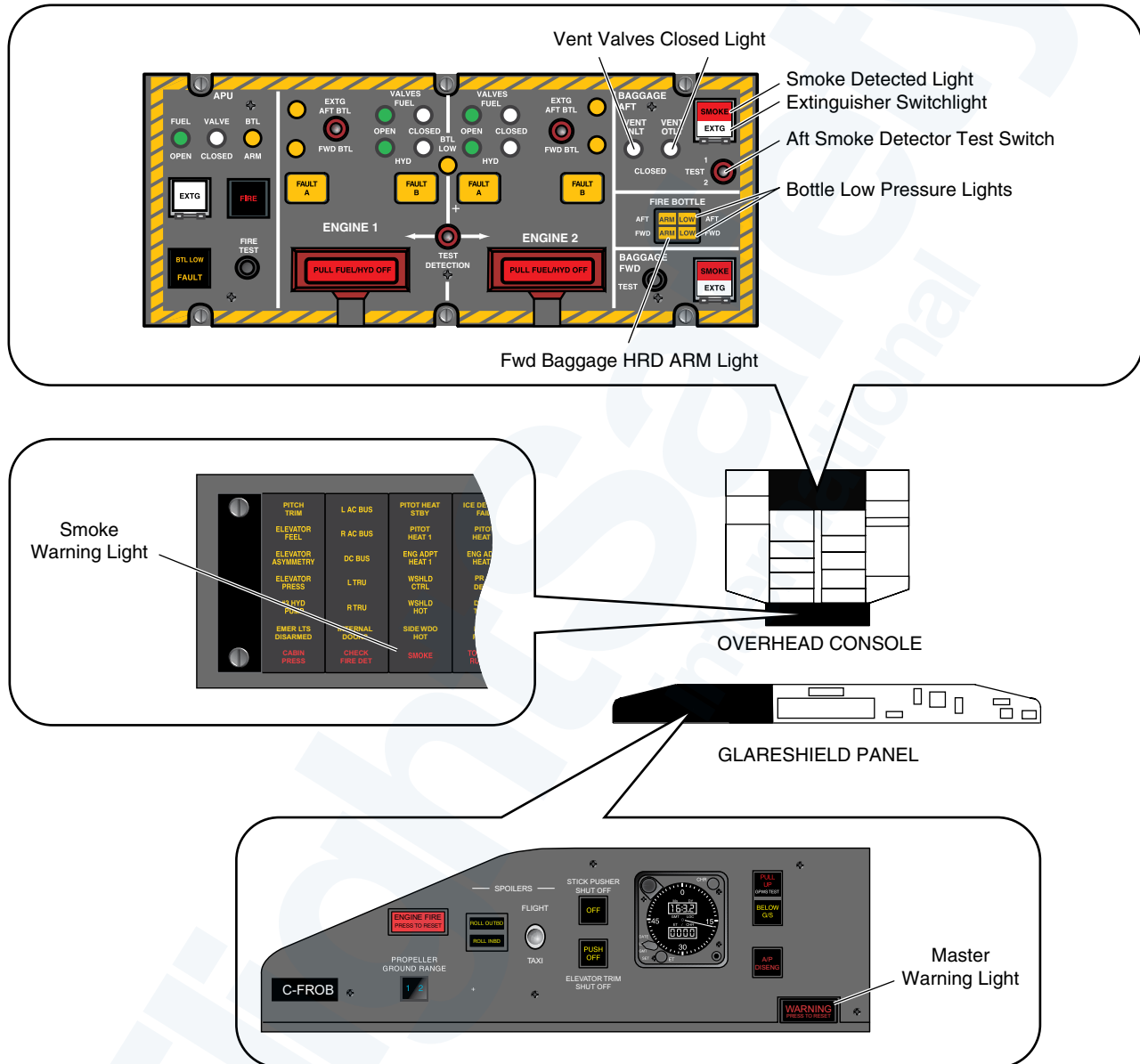
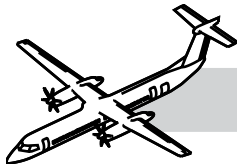
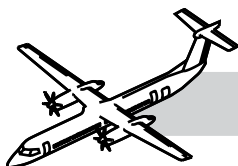


Figure 26-33. Fire Protection Panel Aft Baggage Smoke Detection



COMPONENT DESCRIPTION

Aft High Rate Fire Extinguisher Bottle

Refer to Figure 26-34. Aft Baggage High Rate Fire Bottle and Cartridge.

The aft baggage compartment HRD bottle is installed at the LH side in the aft fuselage aft of the LRD bottle.

Electrical connections are provided for the squib and the bottle monitor pressure switch.

The HRD bottle is a stainless steel container filled with Halon 1301 and pressurized with Nitrogen gas.

The HRD bottle has one discharge valve actuated by a cartridge (squib) in a housing assembly with an electrical connection. The bottle uses a dual-bridgewire electro-explosive device (EED) within the cartridge.

Thermal relief is provided through a burst disc and then the relief is through the distribution system into the respective compartment.

NOTE

Use of Halon 1301 does not require maintenance clean up after a discharge into the baggage compartment. This light will come on when the AFT HRD bottle has lost pressure.

A manual exercise key access in the pressure switch is supplied for ground check of proper pressure switch operation. The key will open the pressure switch contacts, to simulate a depressurized bottle condition. The HRD bottle pressure switch status is indicated by the AFT "BTL LOW" light on the FPP.

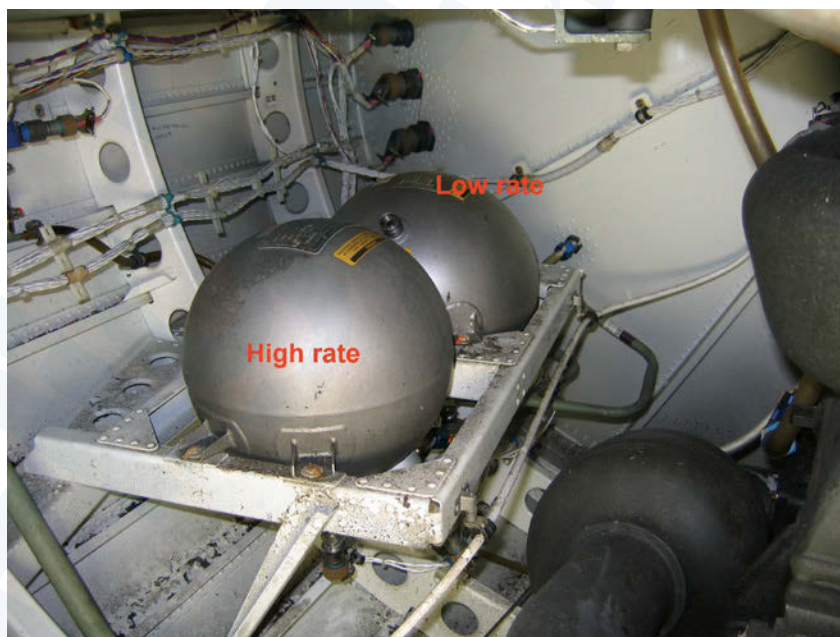
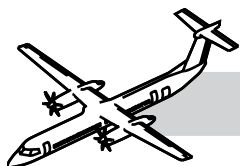


Figure 26-34. Aft Baggage High Rate Fire Bottle and Cartridge

**LEGEND**

1. Bracket
2. Clamps
3. Discharge Valve
4. Cartridge (Squib)
5. Discharge Tube
6. Lanyard
7. Electrical Connector
8. Pressure Switch/Gauge

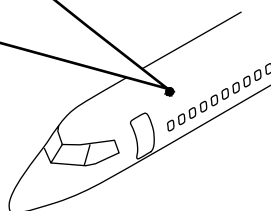
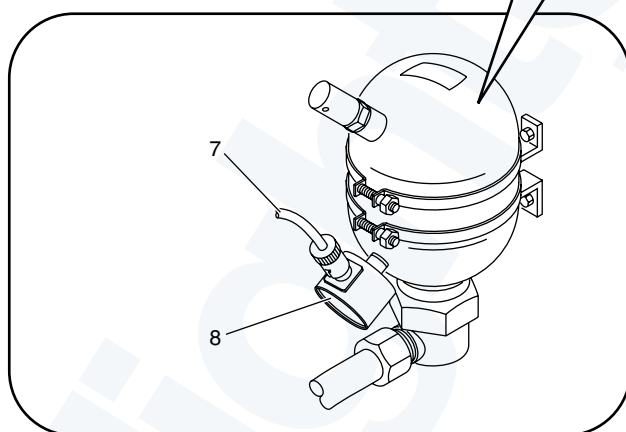
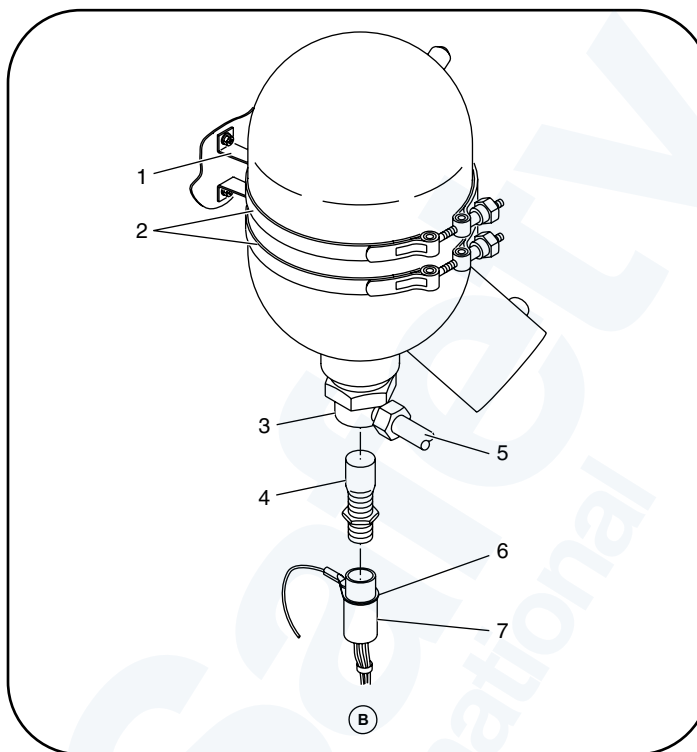
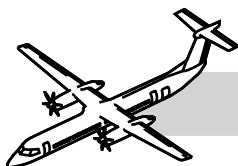


Figure 26-35. Forward High Rate Fire Extinguisher Bottle and Cartridge



Fwd High Rate Fire Extinguisher Bottle

NOTES

Refer to Figure 26-35. Forward High Rate Fire Extinguisher Bottle and Cartridge.

There is one High Rate fire extinguisher bottle for the forward baggage compartment. The bottle is installed on the forward wall of the forward baggage compartment. The bottle container is attached with band clamps to brackets on the wall. Electrical connections are provided to the discharge valve with the cartridge (squib) and to the bottle monitor pressure gauge and switch.

The High Rate fire extinguisher bottle is a stainless steel container charged with Halon 1301 and pressurized Nitrogen.

Post Mod fire bottles have a gauge. Pre Mod bottles without a gauge will have a manual exercise key for ground check of proper pressure switch operation. The HRD bottle pressure switch status is indicated by the Aft BTL LOW advisory light on the FPP. This light illuminates when the AFT HRD bottle has lost pressure.

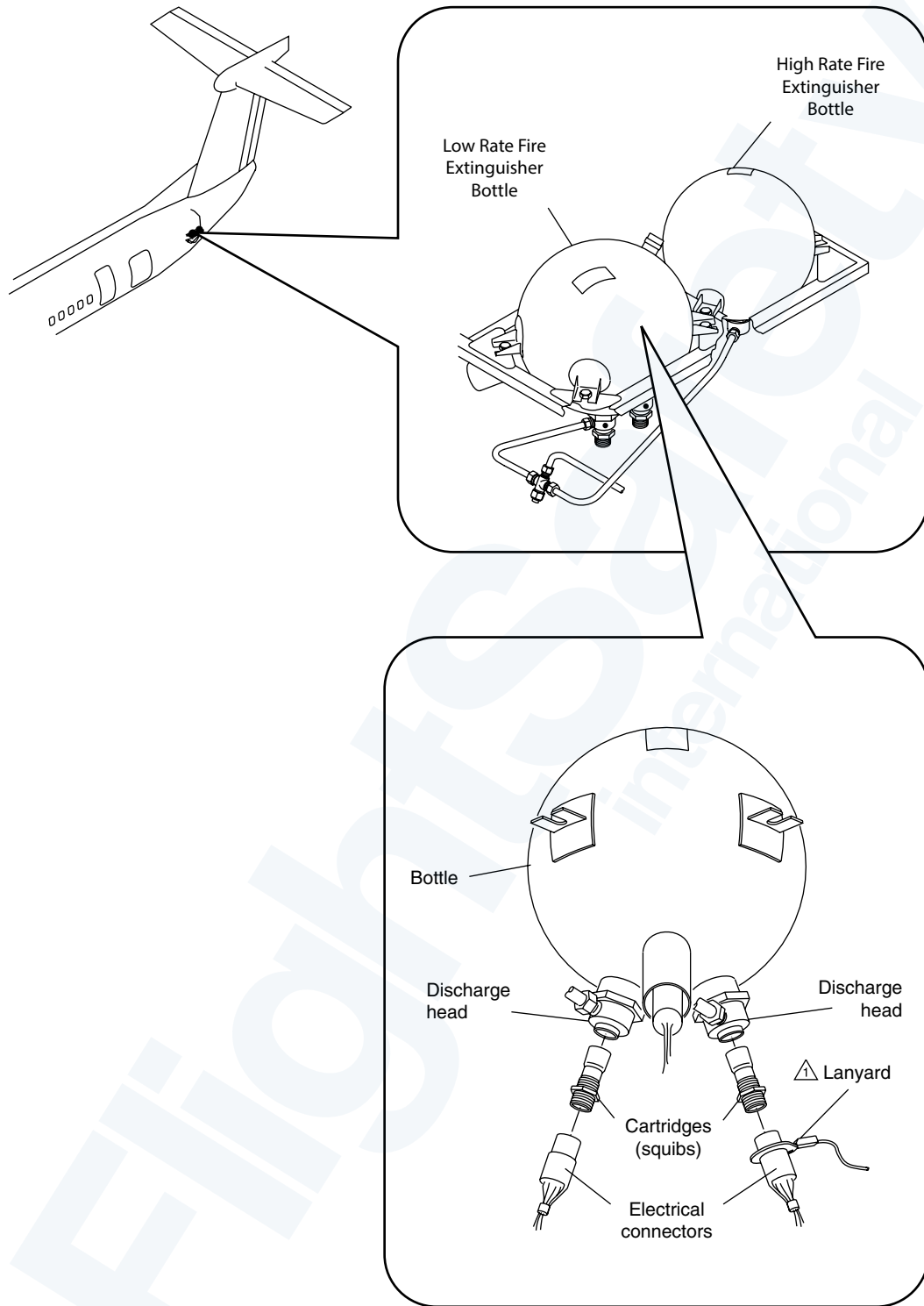
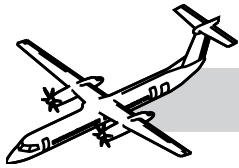
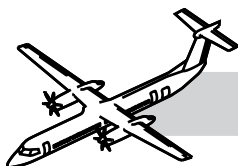


Figure 26-36. Baggage Low Rate Fire Extinguisher Bottles and Cartridge



Low Rate Fire Extinguisher Bottle

NOTES

Refer to Figure 26-36. Baggage Low Rate Fire Extinguisher Bottles and Cartridge.

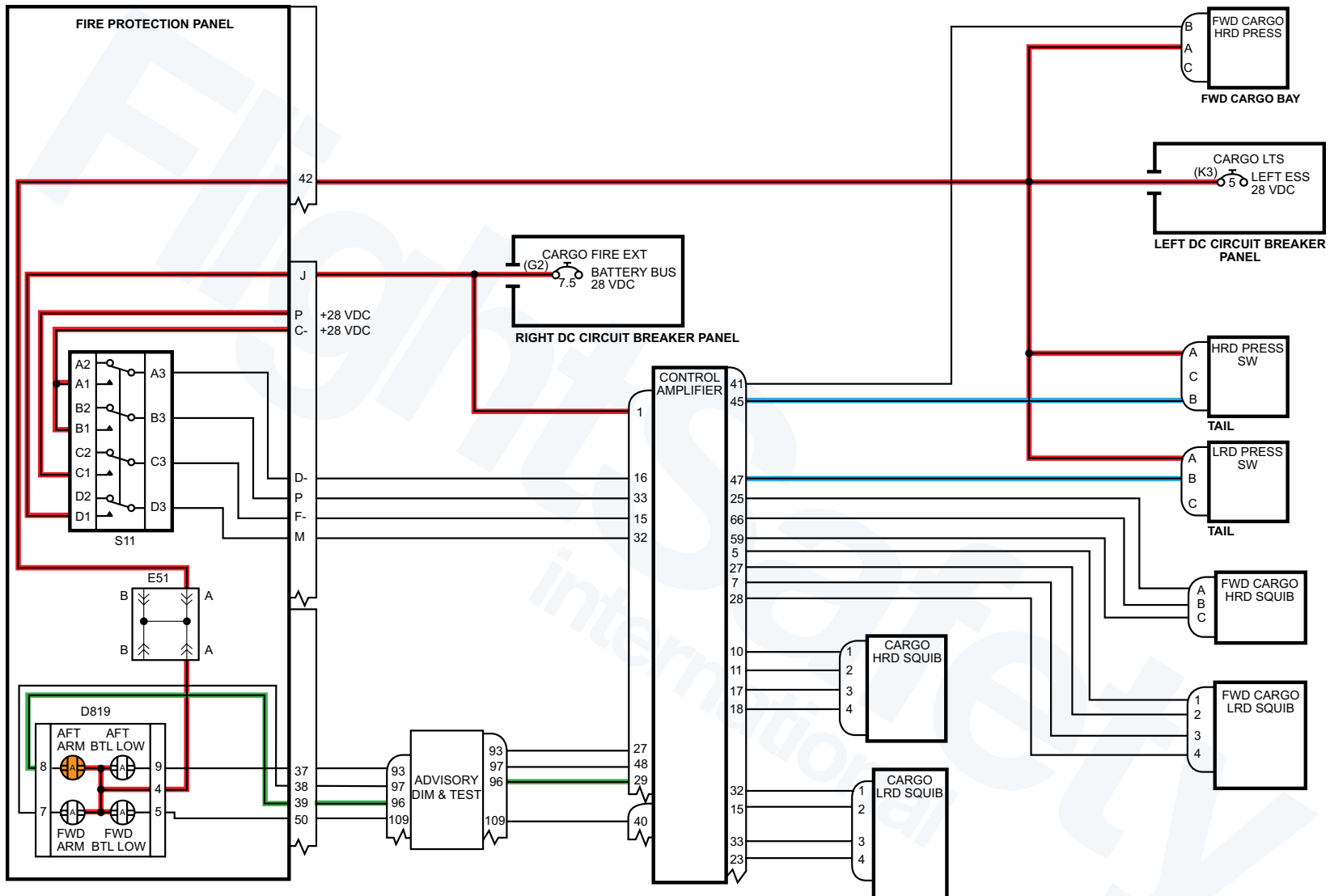
The baggage compartment LRD bottle is forward of the Aft HRD bottle at the LH side of the aft fuselage in the aft equipment bay.

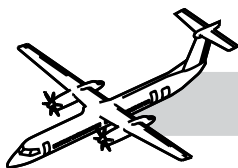
Electrical connections are provided for the squib and the bottle monitor pressure switch.

The LRD bottle is filled with Halon 1301 and pressurized with Nitrogen gas.

A metering device inside the low rate fire extinguisher controls the slow release of the extinguishing agent from the bottle. This device lets the bottle discharge over a period of 15 minutes.

The baggage compartment will retain a minimum of 3% by volume concentration of Halon for a minimum of 45 minutes due to the construction of the compartment and the discharge arrangements of the extinguishers.





OPERATION

Extinguish (AFT)

Figure 26-37. Aft Baggage Smoke Detected.

Pressure in the AFT HRD bottle and LRD bottle is monitored by the control amplifier. The pressure switches receive 28 VDC from the left essential bus via the Cargo LTS circuit breaker K3.

The continuity of the squibs is also monitored. If both the HRD bottle pressure switch shows pressure and the squib shows continuity when smoke is detected, the control amplifier will output a ground to light the appropriate ARM light.

Refer to Figure 26-38. Fire Protection Panel Detail Aft Baggage Smoke.

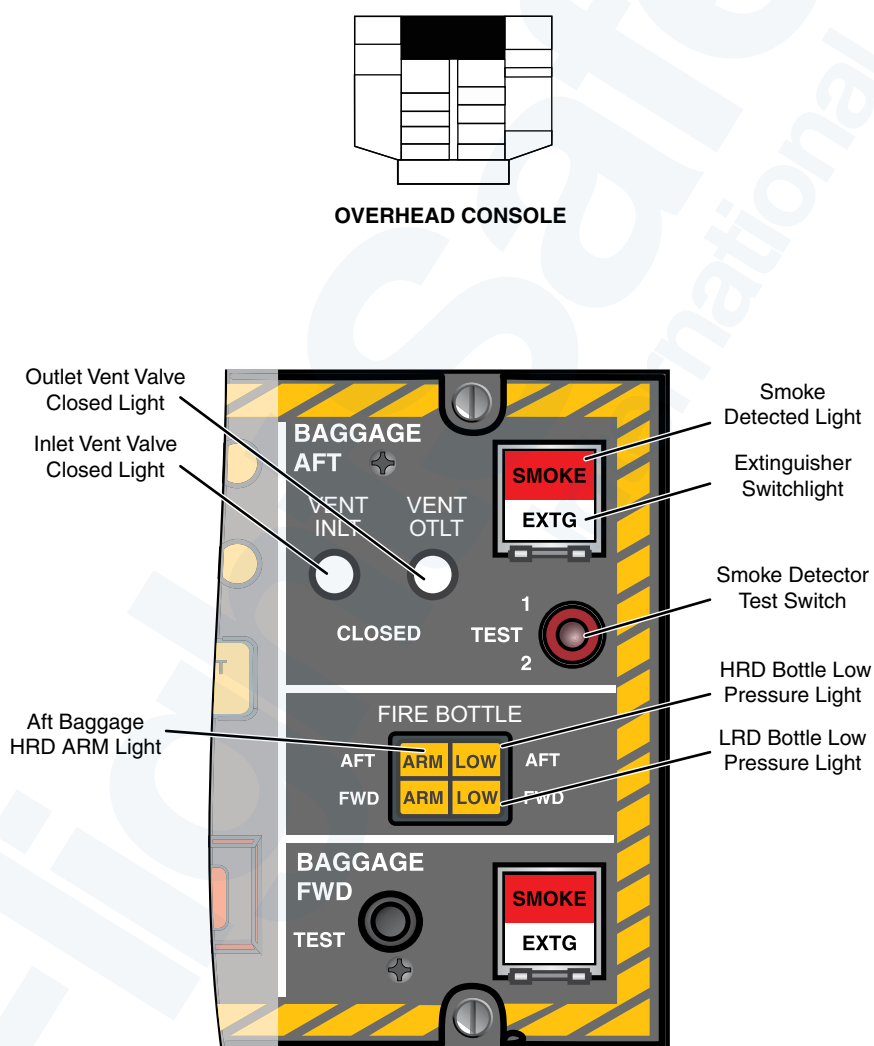


Figure 26-38. Fire Protection Panel Detail Aft Baggage Smoke

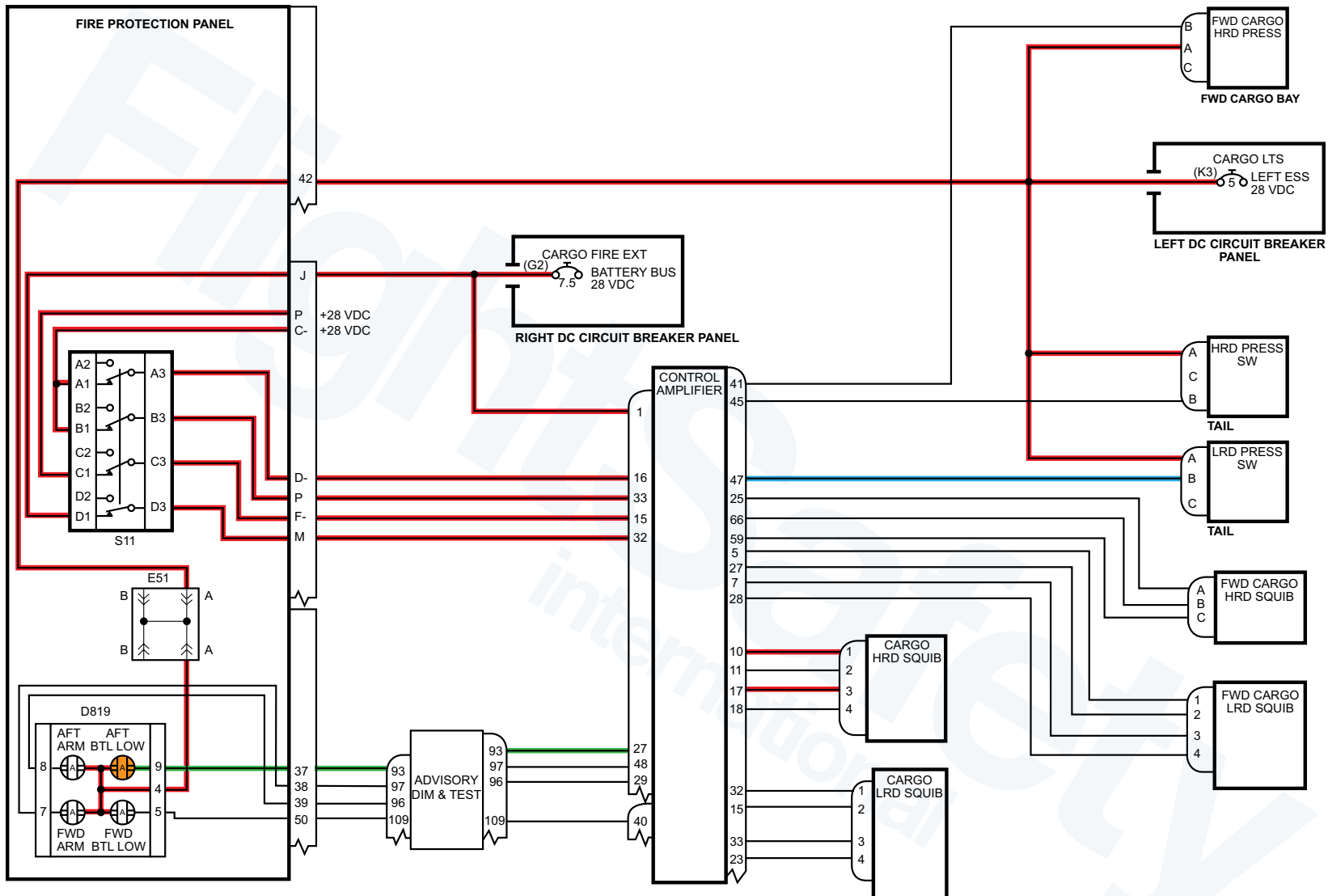


Figure 26-39. Aft Baggage Fire Extinguished (Sheet 1 of 2)

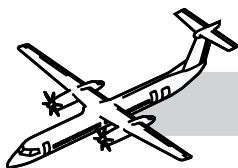


Figure 26-39. Aft Baggage Fire Extinguished
(Sheet 1 of 2).

NOTES

When the extinguisher switch is pushed, 28 VDC is supplied to the control amplifier pins 15, 16, 32 and 33.

The control amplifier outputs 28 VDC to the AFT HRD bottle on 2 separate lines and the cartridge fires. The HRD bottle pressure goes low, therefore, the armed light extinguishes and the BTL LOW light illuminates.

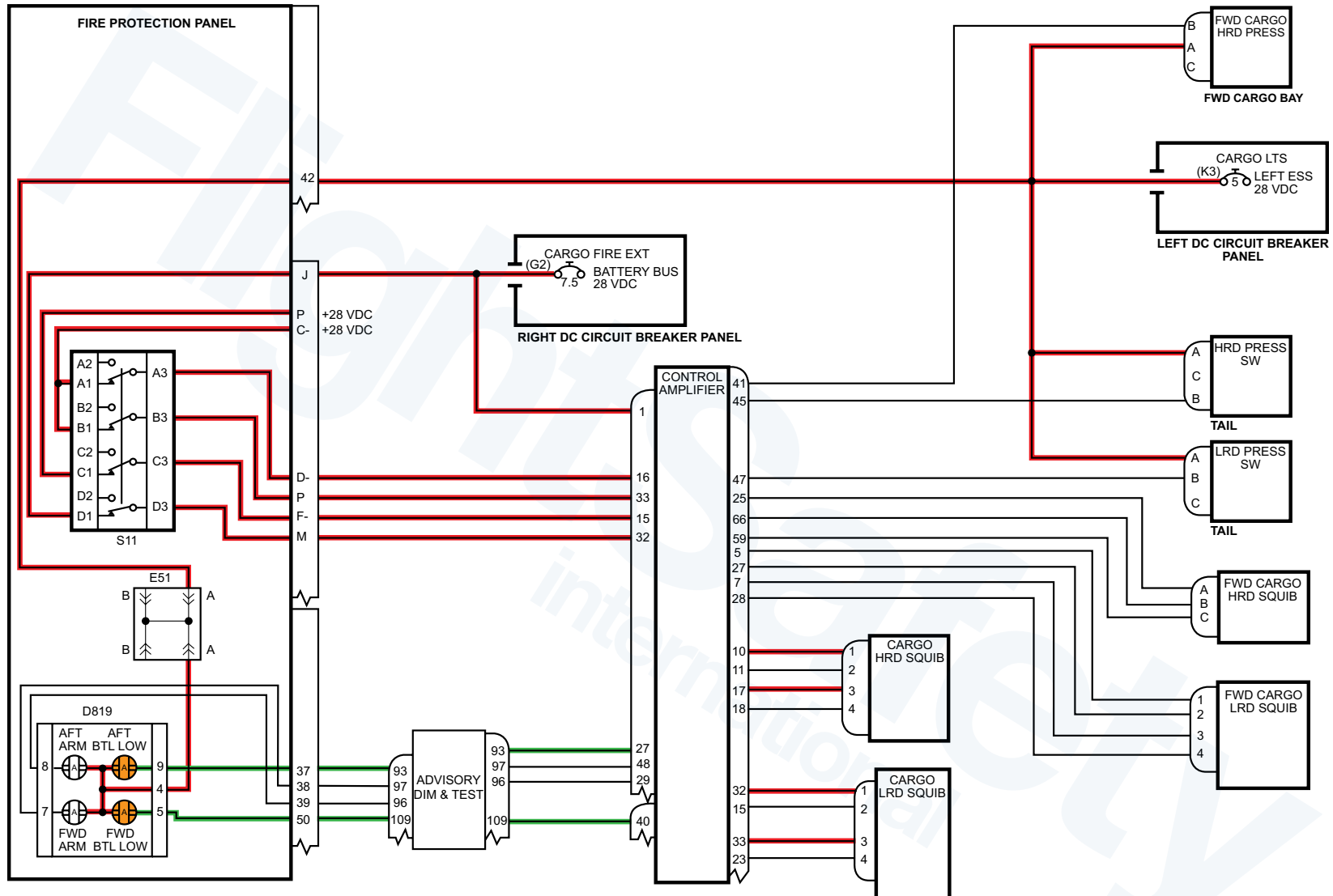
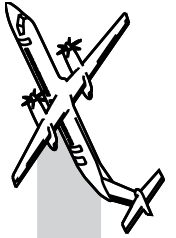
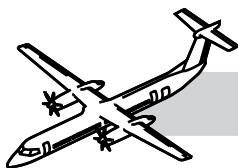


Figure 26-40. Aft Baggage Fire Extinguished (Sheet 2 of 2)



Refer to Figure 26-40. Aft Baggage Fire Extinguished (Sheet 2 of 2).

NOTES

Seven minutes after the extinguish switch is pushed, the control amplifier outputs 28 VDC to the AFT cargo squib on the LRD bottle. Due to flow restriction, it takes 15 minutes before the LRD pressure switch will show low pressure. At this time the FWD BTL LOW advisory light will illuminate.

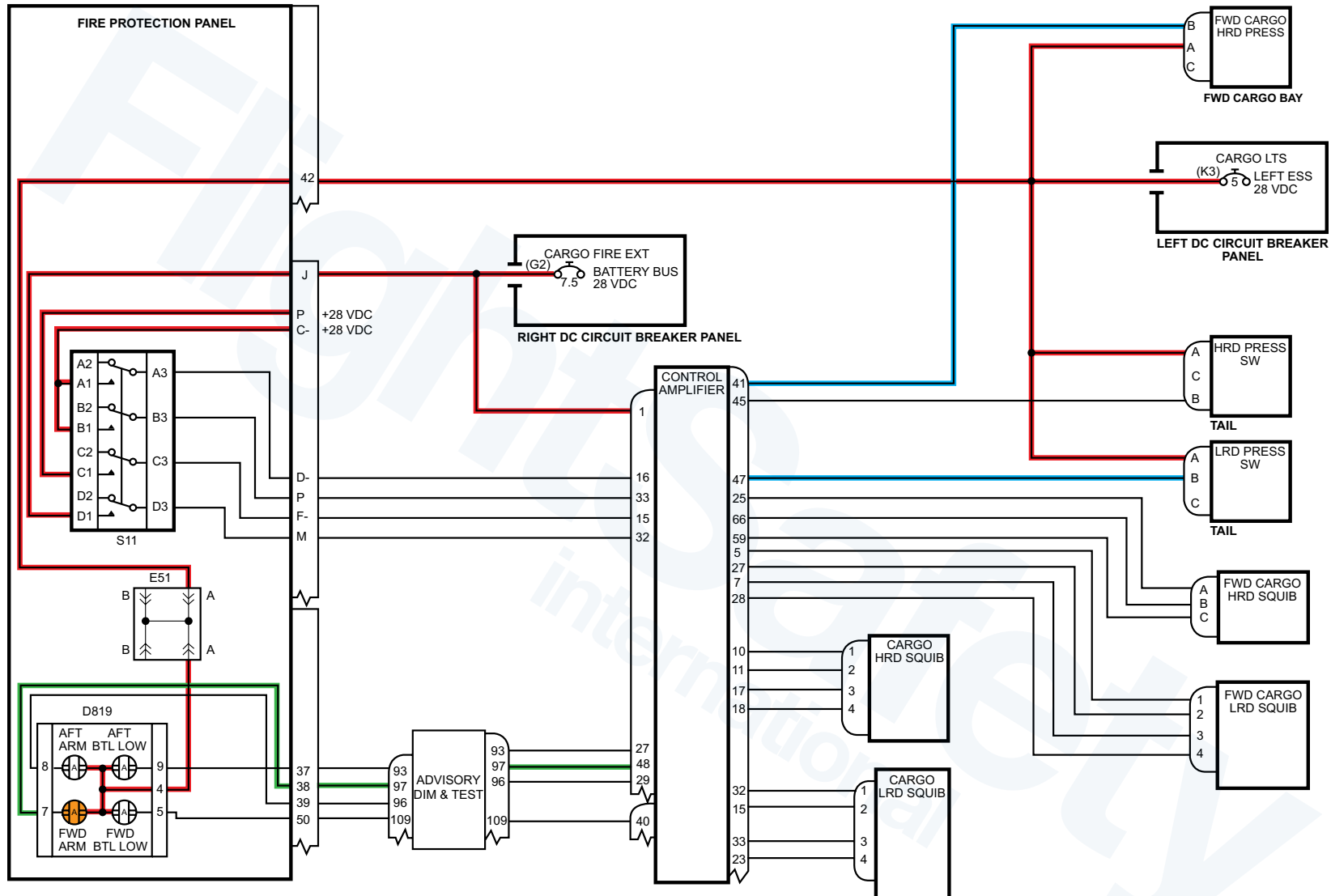
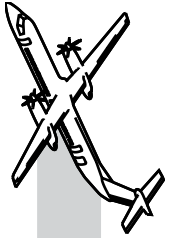
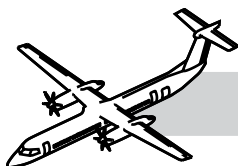


Figure 26-41. Forward Baggage Compartment Fire Detected



Extinguish (FWD)

NOTES

Refer to Figure 26-41. Forward Baggage Compartment Fire Detected.

Pressure in the FWD HRD bottle and LRD bottle is monitored by the control amplifier. The pressure switches receive 28 VDC from the left essential bus via the Cargo LTS circuit breaker K3.

The continuity of the squibs is also monitored. If both the FWD bottle pressure switch shows pressure and the squib shows continuity when smoke is detected, the control amplifier will output a ground to light the appropriate ARM light.

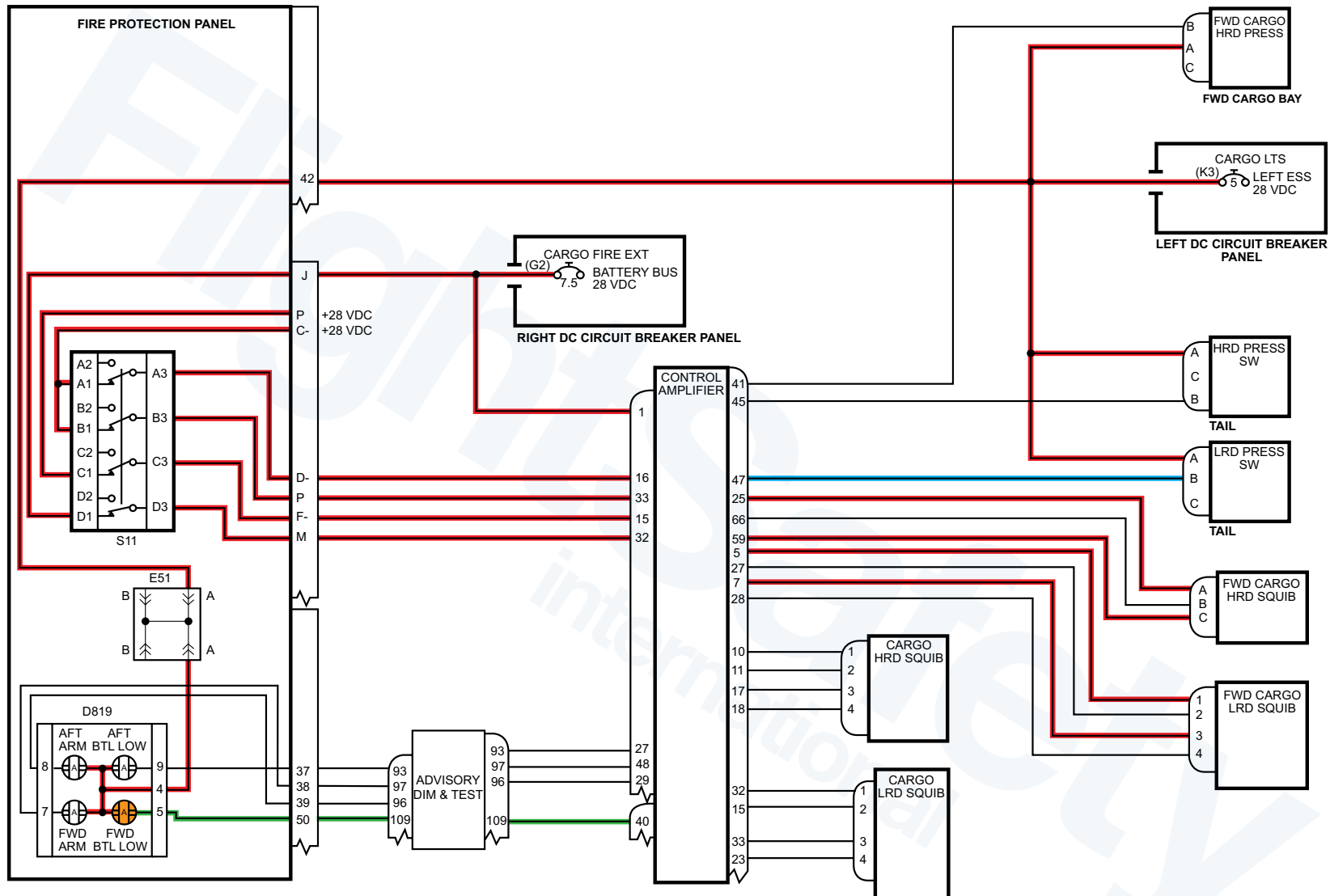
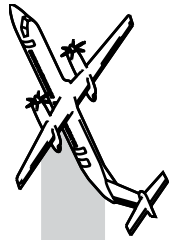


Figure 26-42. Forward Baggage Compartment Fire Extinguished (Sheet 1 of 2)

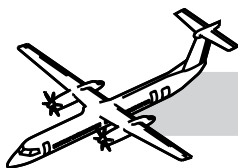


Figure 26-42. Forward Baggage Compartment
Fire Extinguished (Sheet 1 of 2).

NOTES

When the extinguisher switch is pushed, 28 VDC is supplied to the control amplifier pins 15, 16, 32 and 33.

The control amplifier outputs 28 VDC to the FWD HRD and the LRD bottles on 2 independent lines and the cartridges fire. The HRD bottle pressure goes low, therefore, the armed light extinguishes and the BTL LOW light illuminates.

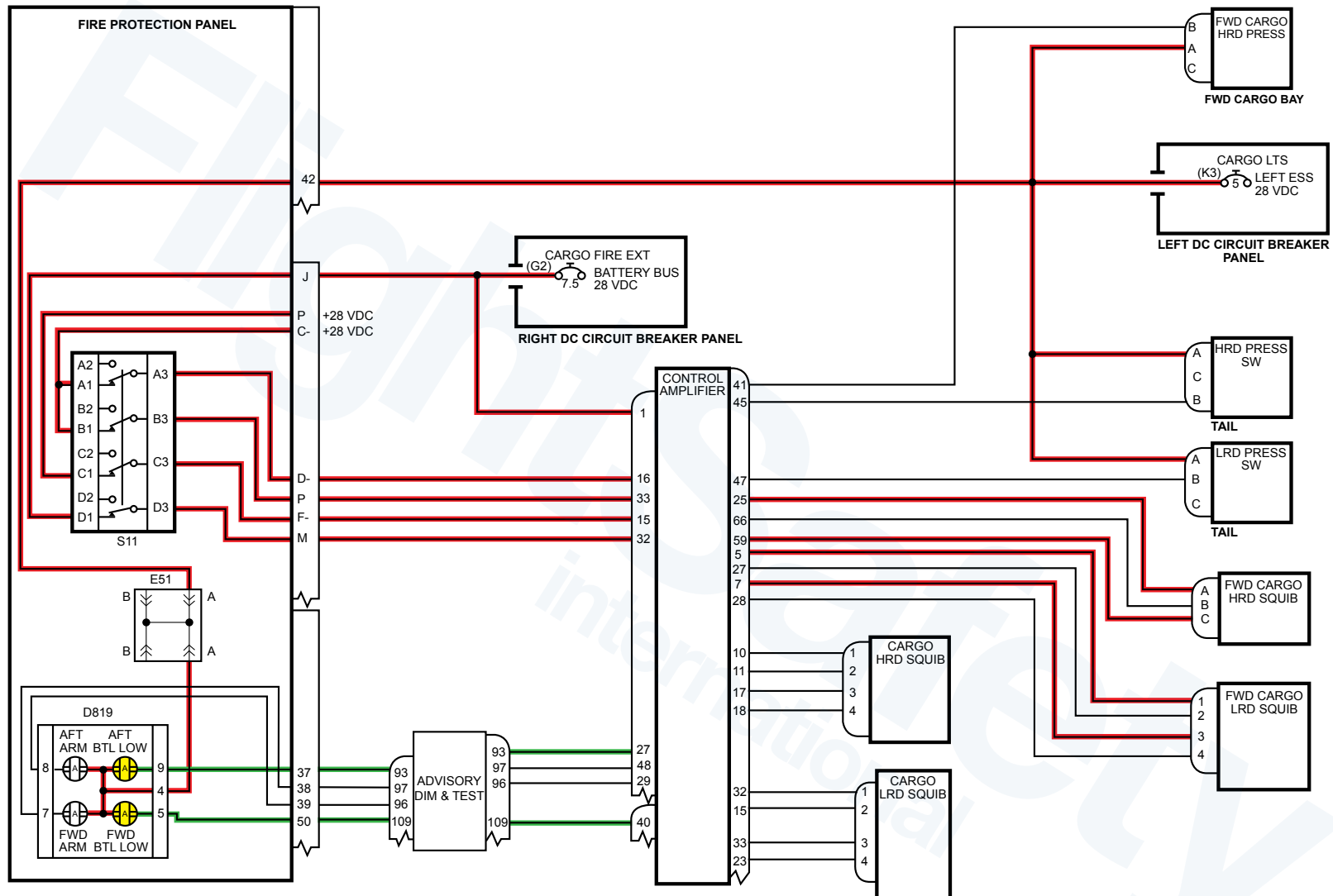
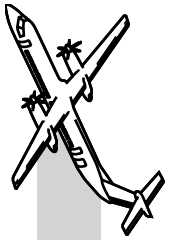
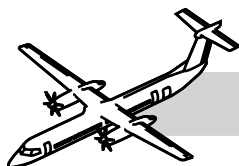


Figure 26-43. Forward Baggage Compartment Fire Extinguish (Sheet 2 of 2)



Refer to:

- Figure 26-43. Forward Baggage Compartment Fire Extinguish (Sheet 2 of 2).
- Figure 26-44. Fire Protection Panel Fwd Baggage Smoke Detection.

The HRD and LRD bottles fire at the same time. Due to restrictors, the LRD bottle pressure switch will not activate until 15 minutes after squib discharge. At this time the AFT BTL LOW advisory light illuminates.

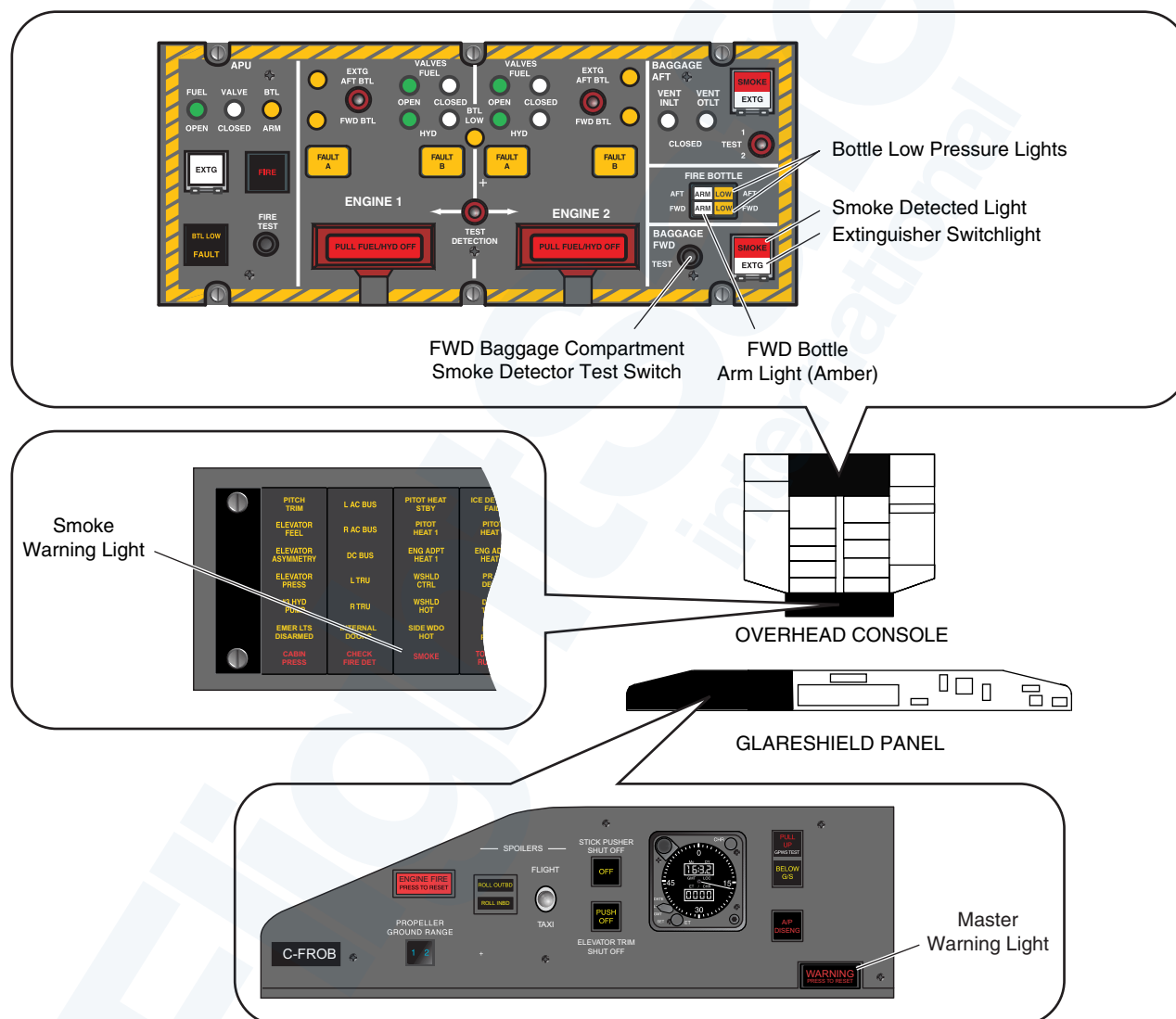
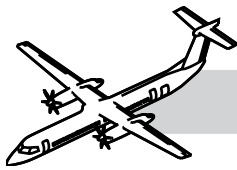


Figure 26-44. Fire Protection Panel Fwd Baggage Smoke Detection



Lavatory Smoke Detection

The lavatory smoke detector alerts the cabin crew of smoke in the lavatory.

Refer to Figure 26-45. Cabin Repeater Lights.

Smoke detected in the lavatory will cause a local red detector Light Emitting Diode (LED) to come on and a local audio alert to sound.

Smoke detected also causes the passenger compartment ceiling repeater red lights to illuminate and a single audible warning (high chime) in the passenger address system to sound.



Figure 26-45. Cabin Repeater Lights

Refer to Figure 26-46. Lavatory Smoke Detector.

The lavatory smoke detector has the following:

A green LED: Indicates the smoke detector is powered by the aircraft DC bus.

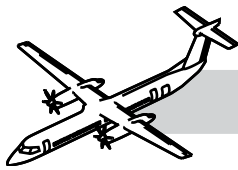
A red alarm LED and aural alert: Indicates smoke has been detected.

A test switch is used to test the smoke detector. A successful test will bring on the red alarm LED and the local aural alert.

An alarm interrupt switch is used to silence the local alarm if it is activated by smoke detection.



Figure 26-46. Lavatory Smoke Detector



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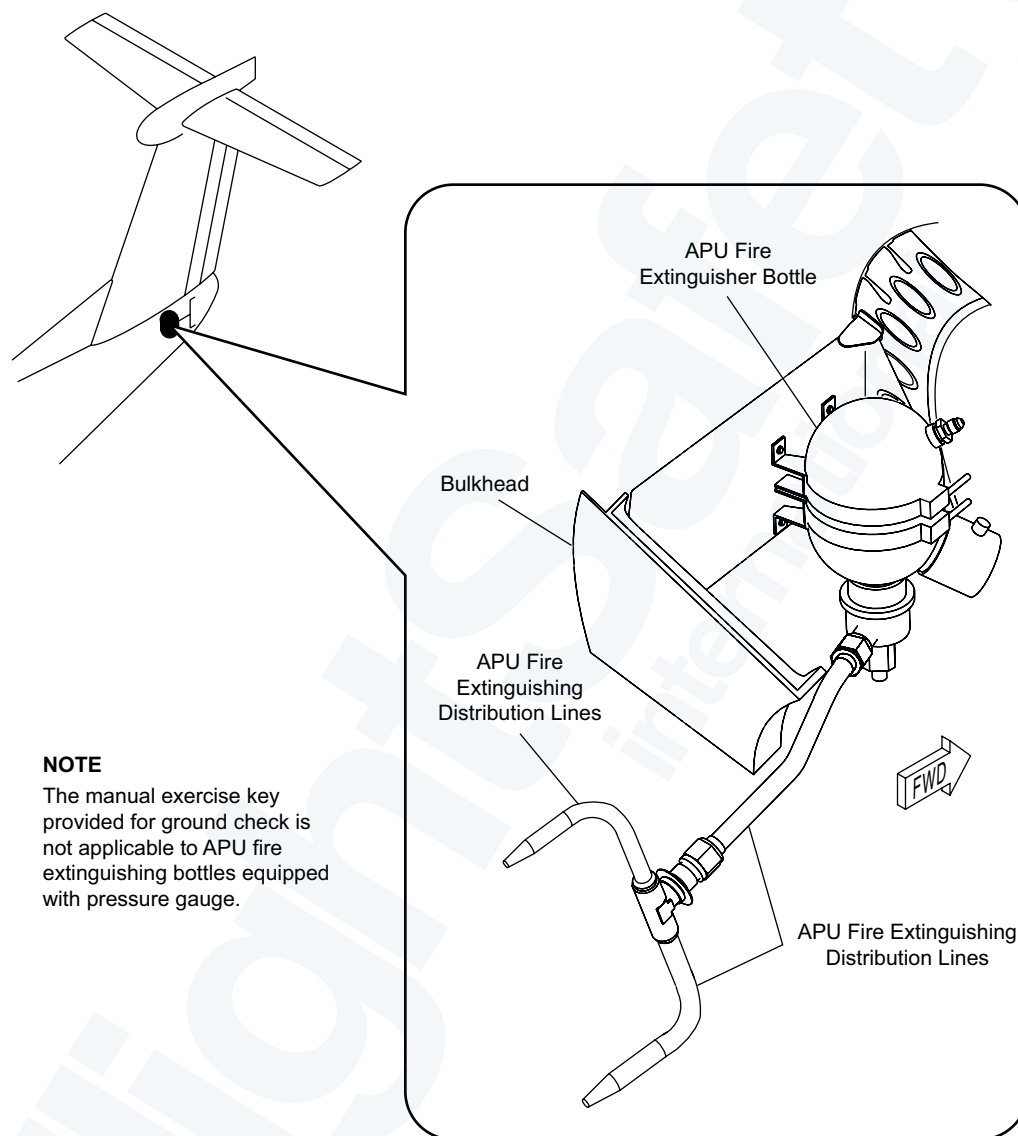
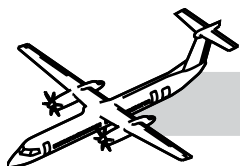
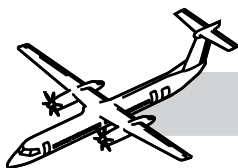


Figure 26-47. APU Fire Extinguishing



26-13-00 APU FIRE DETECTION SYSTEM

INTRODUCTION

APU fire detection system senses fire or overheat conditions in the APU compartment.

GENERAL

Refer to:

- Figure 26-47. APU Fire Extinguishing.
- Figure 26-48. APU Fire Extinguisher.

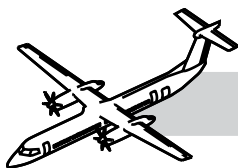
The APU fire detection is by a single APD in the APU compartment. The APD operation is the same as previously described in the engine nacelle section.

The APD interfaces with the control amplifier and the FPP. The fire protection system has a stainless steel fire extinguisher bottle in the aft equipment section.



Figure 26-48. APU Fire Extinguisher





OPERATION

APU Fire Detection System

Refer to Figure 26-49. APU Fire Detection and Extinguishing - Schematic.

The control amplifier interfaces with the FPP and the APU APD. Further it performs detector and suppressor monitoring and test functions, and drives the appropriate lights on the FPP and the CAWP.

When a fire or overheat is sensed in the APU, the control amplifier performs the following:

- FIRE (red) light on the FPP illuminates.
- BTL ARM (amber) light on the FPP illuminates.
- FUEL OPEN (green) advisory light on the FPP extinguishes.
- FUEL CLOSED (white) light on the FPP illuminates.
- EXTG (white) switch light on the FPP illuminates.
- CHECK FIRE DET (red) warning light on the CAWP and flashes.
- Master WARNING PRESS TO RESET (red) light on the Glareshield Panel comes on and flashes.

When a fire or overheat is sensed, the APU will automatically shut down and the extinguisher bottle will fire. The FAIL advisory light on the APU control panel will come on.

The 28 VDC power required for the cartridge passes through two (2) circuit breakers, wired in such a manner that each breaker provides protection for a single shot into the APU fire zone (tailcone).

No pilot action is required to operate APU fire protection. The control amplifier controls a relay which closes the APU fuel valve. When electrical power from the relay is supplied to the Full-Authority Digital-Electronic-Control (FADEC), the **APU FUEL VALVE CLOSED** advisory light on the FPP illuminates.

The APU is shutdown automatically and the FAIL advisory light in the APU control panel illuminates. After a seven seconds delay, an electrical signal from the control amplifier ignites an Electro-Explosive Device (EED) in the extinguisher cartridge.

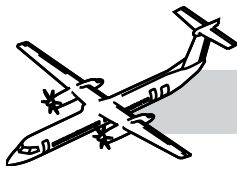
The EED then explodes and the fire extinguishing agent is automatically released through the distribution tubes to the APU compartment. The **APU BTL ARM** light on the FPP extinguishes and the **BTL LOW** light illuminates.

If automatic fire extinguishing fails and a fire or overheat condition is still sensed, a manual override capability is available by operating the APU EXTG switch on the FPP.

APU fire extinguisher bottle pressure is shown by a **BTL LOW** light on the FPP. The APU will automatically shut down.

Manual operation also causes operation of the circuit which closes the APU fuel shutoff valve, which is indicated by the **APU FUEL VALVE CLOSED** advisory light illuminating.

The APU fire bottle is constantly monitored by the control amplifier for pressure during a non-alarm state. The bottle pressure switch status is indicated by the **BTL LOW** light on the FPP. The light will come on when the APU fire bottle has lost pressure. The **APU BTL ARM** light will come on during the APU FIRE TEST, if at least one squib bridgewire has continuity and power.



OPERATIONAL TEST OF THE APU FIRE DETECTION SYSTEM

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

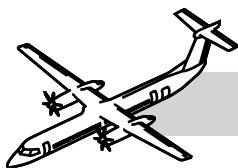
Obey all electrical/electronic safety precautions.

Make sure the aircraft electrical system is energized.

Make sure the CB's listed in the Task sheet are CLOSED.

Do an Operational Test of the APU fire detection system as follows:

- Select APU Power switch ON
- Observe APU FADEC BIT
- On the FPP check APU FUEL VALVE OPEN light is ON
- Check APU FUEL VALVE CLOSED light is OFF
- Push and hold APU FIRE TEST pushbutton. Make sure that:
 - The APU FIRE indicator light is ON (FPP)
 - The BTL ARM indicator light is ON (FPP)
 - The APU EXTG switchlight is ON (FPP)
 - The APU VALVE CLOSED light is ON (FPP)
 - The FAULT indication is ON (APU control panel)
 - The FAIL indicator light is ON (APU control panel)
 - The APU caution light is ON
 - The CHECK FIRE DET warning light is ON
- The MASTER WARNING and MASTER CAUTION lights are ON.
- Release APU FIRE TEST BUTTON. Make sure that:
 - APU FUEL VALVE OPEN indicator light goes OFF
 - APU FUEL VALVE CLOSED light comes ON
 - The APU FIRE indicator light is OFF (FPP)
 - The BTL ARM indicator light is OFF (FPP)
 - The APU EXTG switchlight is OFF (FPP)
 - The FAULT indication is OFF (APU control panel)
 - The FAIL indicator light is OFF (APU control panel)
 - The APU caution light is OFF
 - The CHECK FIRE DET warning light is OFF
 - The MASTER WARNING and MASTER CAUTION lights are OFF.
- Push APU PWR switch to remove the power.



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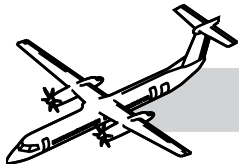
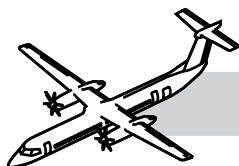


Figure 26-50. Lavatory Fire Extinguisher and Waste-Bin



26-24-00 LAVATORY FIRE EXTINGUISHING

INTRODUCTION

Refer to Figure 26-50. Lavatory Fire Extinguisher and Waste-Bin.

SYSTEM DESCRIPTION

Refer to Figure 26-51. Lavatory Compartment Waste-Bin Fire Extinguisher.

The lavatory fire extinguisher has dual-discharge outlets that are thermally actuated with no electrical interface. When a fire occurs in the lavatory compartment waste-bin and the temperature reaches 174°F (79°C), it causes fusible seals to melt and release the end caps from the discharge tubes. The extinguishing agent is then released and discharged into the bin.

The lavatory fire extinguisher cannot be refilled or reused. It requires periodic weighing to make sure that it is full. If the discharge tubes are bent beyond the specified angle, fire suppression may not work properly.

The lavatory fire extinguisher uses Halon 1211 as the extinguishing agent.

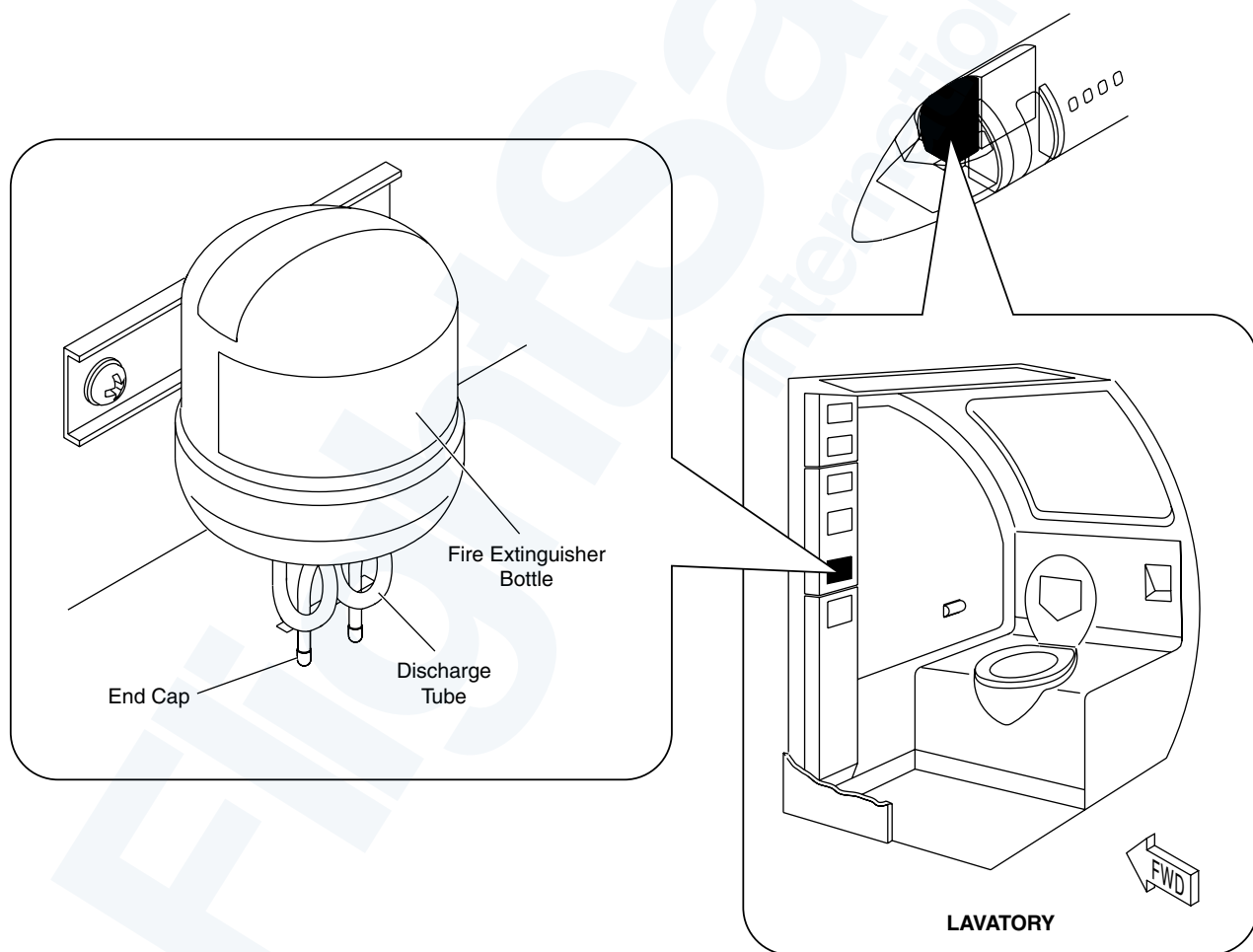


Figure 26-51. Lavatory Compartment Waste-Bin Fire Extinguisher

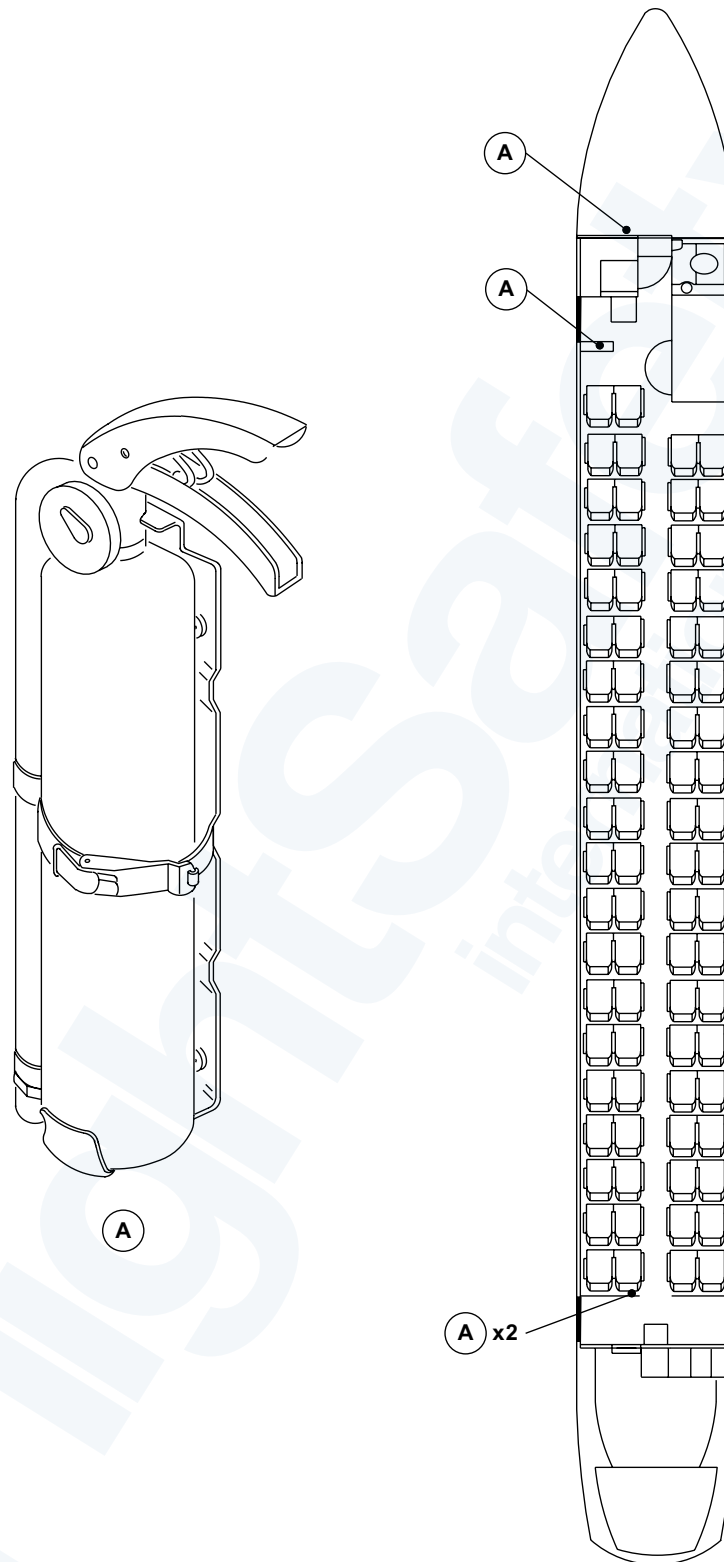
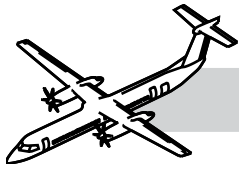
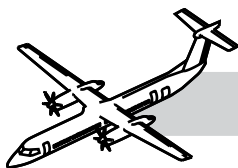


Figure 26-52. Hand Held Fire Extinguishers



26-25-00 PORTABLE HAND OPERATED FIRE EXTINGUISHERS

COMPONENT DESCRIPTION

Flight Compartment Fire Extinguisher and Passenger Cabin Fire Extinguishers

Refer to Figure 26-52. Hand Held Fire Extinguishers.

Hand Operated Fire Extinguishers

Refer to:

- Figure 26-53. Hand Operated Flight Compartment Fire Extinguisher.
- Figure 26-54. Aft Cabin Fire Extinguishers.
- Figure 26-55. Hand Operated Fire Extinguisher - Aft Passenger Compartment.
- Figure 26-56. Hand Operated Fire Extinguisher - Fwd Passenger Compartment.

The portable hand-operated fire extinguishers are filled with Halon 1211. Each bottle is suitable for use on electrical, fuel or oil fires. Halon 1211 is not toxic or corrosive. It does not cause cold burns, harm fabrics or metals and does not leave residue on electrical components.

Flight Compartment Fire Extinguisher

Refer to Figure 26-53. Hand Operated Flight Compartment Fire Extinguisher.

A single portable hand operated fire extinguisher in the flight compartment is attached to the left bulkhead behind the pilot's seat. Instructions for use of the extinguisher are shown on the body of the extinguisher bottle.

Passenger Cabin Fire Extinguishers

Refer to:

- Figure 26-54. Aft Cabin Fire Extinguishers.
- Figure 26-55. Hand Operated Fire Extinguisher - Aft Passenger Compartment.
- Figure 26-56. Hand Operated Fire Extinguisher - Fwd Passenger Compartment.

The passenger compartment is protected by three portable hand-operated fire extinguishers.

A single portable hand operated fire extinguisher is in the forward draft bulkhead stowage.

Two portable hand operated fire extinguishers are in a drawer attached to the left aft draft bulkhead forward side.

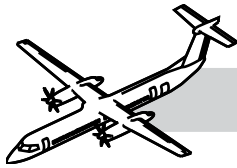


Figure 26-53. Hand Operated Flight Compartment Fire Extinguisher



Figure 26-54. Aft Cabin Fire Extinguishers

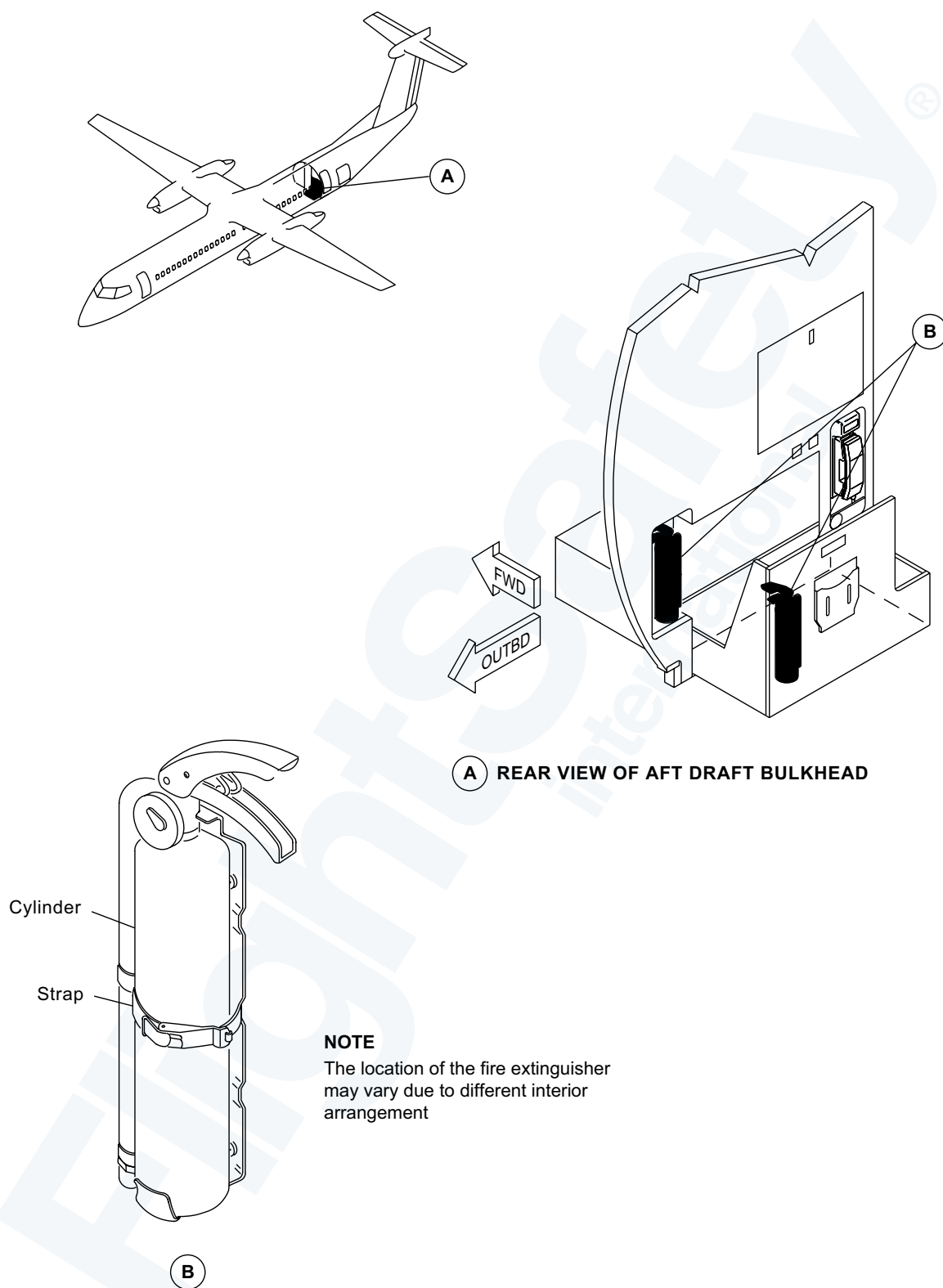
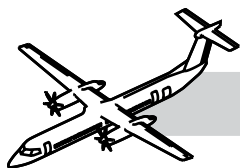


Figure 26-55. Hand Operated Fire Extinguisher - Aft Passenger Compartment

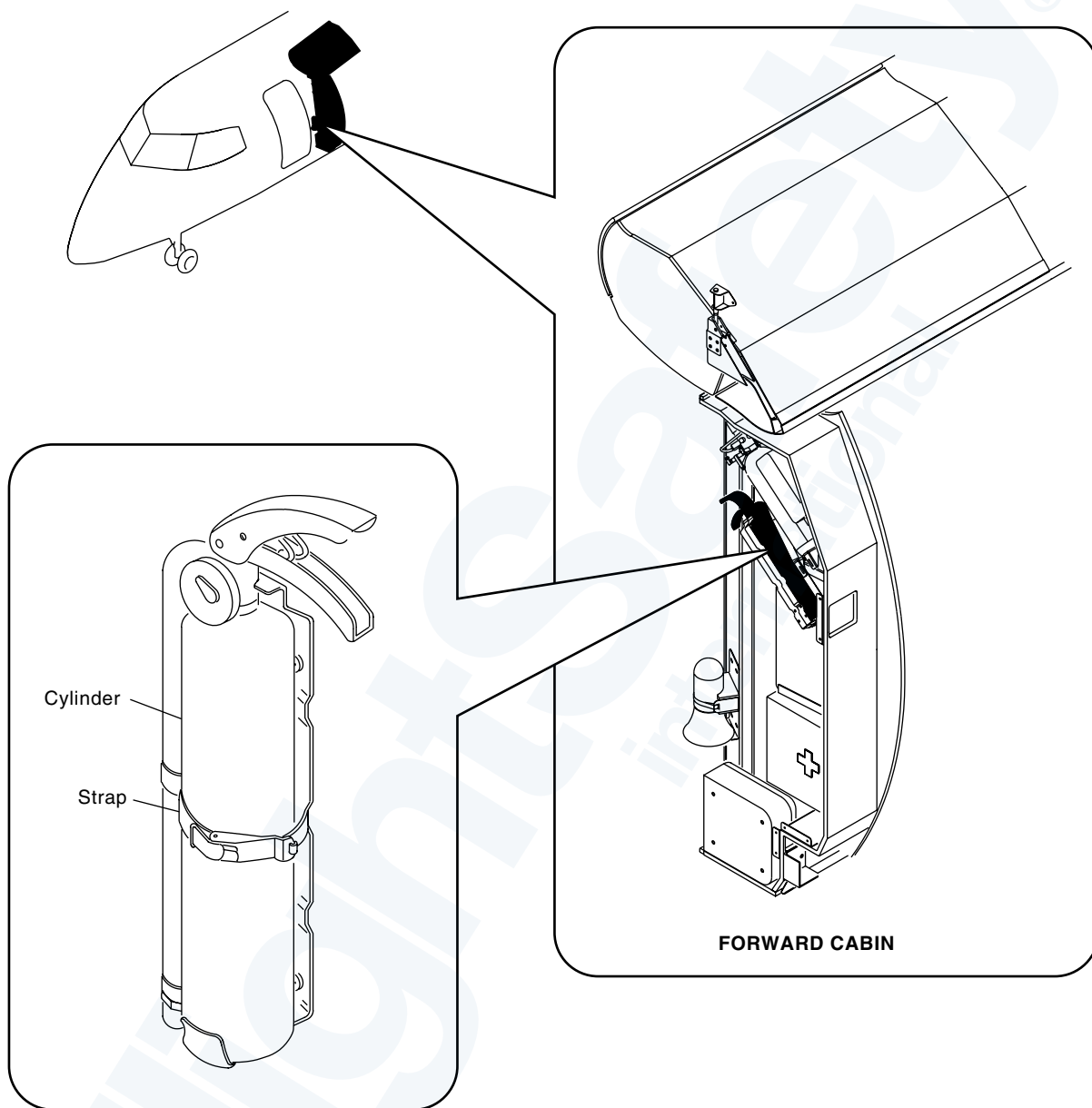
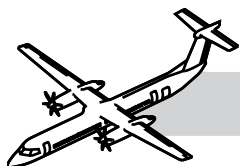
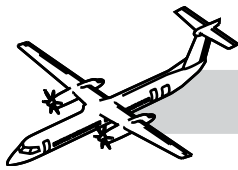
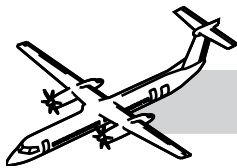


Figure 26-56. Hand Operated Fire Extinguisher - Fwd Passenger Compartment



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26-00-00 APPENDIX

MAINTENANCE CONSIDERATION

Safety Precautions

WARNING

THE DISCHARGE CARTRIDGE (SQUIB) IS AN ELECTRICALLY FIRED EXPLOSIVE DEVICE. WHEN YOU DO MAINTENANCE, MAKE SURE THAT THE PINS OF THE ELECTRICAL CONNECTOR (SQUIB) HAVE THE SAME ELECTRICAL POTENTIAL. INSTALL A SHUNT PLUG ON THE ELECTRICAL CONNECTOR OF THE SQUIB IMMEDIATELY AFTER YOU DISCONNECT THE ELECTRICAL CONNECTOR. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS IF THE SQUIB IS ACCIDENTALLY ENERGIZED.

WARNING

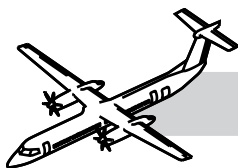
DO NOT MOVE THE EXTG SWITCH TO THE AFT BTL OR THE FWD BTL POSITION. IF YOU DO THIS, YOU WILL RELEASE THE PRESSURE FROM THE FIRE EXTINGUISHER BOTTLE. THIS CAN CAUSE INJURY TO PERSONNEL.

WARNING

BE CAREFUL WITH THE FIRE EXTINGUISHER. THE FIRE EXTINGUISHER IS PRESSURIZED. IT CONTAINS HALON 1301. DO NOT BEND THE DISCHARGE TUBES. IF YOU DO THIS, THE PRESSURE CAN CAUSE INJURY AND THE HALON FUMES ARE POISONOUS.

CAUTION

DO NOT USE MORE THAN 2.5 LBF IN (0.28 NM) OF TORQUE WHEN YOU TURN THE TEST POINT. IF YOU DO NOT DO THIS, YOU CAN CAUSE DAMAGE TO THE PRESSURE SWITCH AND THE FIRE BOTTLE WILL NOT OPERATE CORRECTLY.



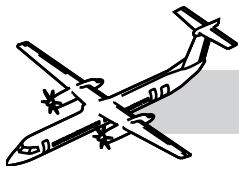
26-00-00 SPECIAL TOOLS & TEST EQUIPMENT

- GSB2400016 Electrical Contact Pin/Socket Kit
- GSB2400001 Digital Multimeter - Hand Held
- Commercially available Shunt Plug
- 473962 Three-wire Firex Squib Circuit Verification Tester (Test Set 1)
- 473959 Four-wire Firex Squib Circuit Verification Tester (Test Set 2)
- GSB2341010 Kit, Two-Way Radio

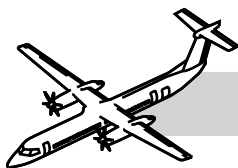
26-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

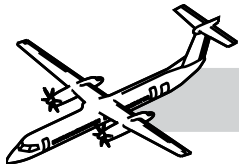
- AMM 26-20-00-710-801: Operational Check of the Fuel and Hydraulic Shut-off Valves (MRB #262000-218).
- FIM 26-10-00-810-801: CHECK FIRE DET (Warning) - Fault Isolation.
- FIM 26-10-00-810-802: FAULT A, ENGINE 1 (Caution) - Fault Isolation.
- FIM 26-10-00-810-803: FAULT B, ENGINE 1 (Caution) - Fault Isolation.
- FIM 26-10-00-810-804: SMOKE (Warning) - Fault Isolation.
- FIM 26-10-00-810-806: FAULT, APU. (Caution) - Fault Isolation.
- FIM 26-10-00-810-807: LOW, AFT FIRE BOTTLE (Caution) - Fault Isolation.
- FIM 26-10-00-810-811: BTL LOW, Engine (Caution) - Fault Isolation.
- FIM 26-10-00-810-812: FAULT A, ENGINE 2 (Caution) - Fault Isolation.
- FIM 26-10-00-810-813: FAULT B, ENGINE 2 (Caution) - Fault Isolation.
- FIM 26-10-00-810-814: LOW, FWD FIRE BOTTLE (Caution) - Fault Isolation.
- FIM 26-10-00-810-815: LOW, FWD and AFT FIRE BOTTLE (Caution) - Fault Isolation.
- FIM 26-10-00-810-816: SMOKE, BAGGAGE AFT (Warning) - Fault Isolation.
- FIM 26-10-00-810-817: SMOKE, BAGGAGE FWD (Warning) - Fault Isolation.
- FIM 26-10-00-810-818: Glareshield panel, an indication discrepancy of the ENGINE FIRE PUSH TO RESET light on the left side - Fault Isolation.
- FIM 26-10-00-810-819: Glareshield panel, an indication discrepancy of the ENGINE FIRE PUSH TO RESET light on the right side - Fault Isolation.
- FIM 26-11-00-810-801: FIRE PROTECTION panel, an indication discrepancy of the FAULT A light in the ENGINE 1 area - Fault Isolation.
- FIM 26-11-00-810-802: FIRE PROTECTION panel, an indication discrepancy of the FAULT B light in the ENGINE 1 area - Fault Isolation.
- FIM 26-11-00-810-803: FIRE PROTECTION panel, an indication discrepancy of the FAULT A light in the ENGINE 2 area - Fault Isolation.



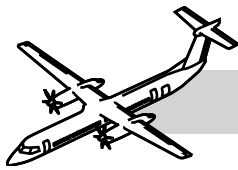
- FIM 26-11-00-810-804: FIRE PROTECTION panel, an indication discrepancy of the FAULT B light in the ENGINE 2 area - Fault Isolation.
- AMM 26-11-00-710-801: Operational Test of the Nacelle Fire Detection System.
- AMM 26-12-06-710-801: Operational Test of the Baggage Smoke Detectors.
- AMM 26-12-11-710-803: Operational Check of the Lavatory Smoke Detector (Kidde Model 3000) (MRB #261000-204).
- FIM 26-13-00-810-801: FIRE, APU area of the FIRE PROTECTION panel (Warning) - Fault Isolation.
- FIM 26-13-00-810-802: FIRE PROTECTION panel, an indication discrepancy of the FAULT light in the APU area - Fault Isolation.
- FIM 26-13-00-810-803: FIRE PROTECTION panel, an indication discrepancy of the FIRE light in the APU area - Fault Isolation.
- AMM 26-13-00-710-801: Operational Test of the APU Fire Detection System.
- AMM 26-20-00-210-801: General Visual Inspection of the LH and RH Engine and APU Fire Extinguishing Distribution Lines and Connections (MRB# 262000-205).
- AMM 26-20-00-210-802: General Visual Inspection of the Forward and Aft Baggage-Compartment LRD and HRD Fire-Extinguishing Distribution-Lines and Connections (MRB#262000-213).
- AMM 26-20-00-210-803: Visual Inspection of the Fire Extinguishing Distribution Tubes (Wing).
- AMM 26-20-00-840-801: Restoration (Hydrostatic Test) of the Nacelle Forward High Rate Discharge (HRD) Fire Bottle (MRB#262000-201).
- AMM 26-20-00-840-802: Restoration (Hydrostatic Test) of the Nacelle Aft High Rate Discharge (HRD) Fire Bottle (MRB#262000-202).
- AMM 26-20-00-840-803: Restoration (Hydrostatic Test) of the Aft Baggage Compartment High Rate Discharge (HRD) Fire Bottle (MRB#262000-206).
- AMM 26-20-00-840-804: Restoration (Hydrostatic Test) of the Forward/Aft Baggage Compartment Low Rate Discharge (LRD) Fire Bottle (MRB#262000-207).
- AMM 26-20-00-840-805: Restoration (Hydrostatic Test) of the Forward Baggage Compartment High Rate Discharge (HRD) Fire Bottle (MRB#262000-210).
- AMM 26-20-00-840-806: Restoration (Hydrostatic Test) of the APU Fire Bottle (MRB#262000-214).
- AMM 26-20-00-900-801: Discard of the Nacelle Forward and Aft High Rate Discharge (HRD) Fire Bottle Cartridges (MRB#262000-203).
- AMM 26-20-00-900-802: Discard of the Aft Baggage Compartment High Rate Discharge (HRD) and the Baggage Compartment(s) Low Rate Discharge (LRD) Fire Bottle Cartridges (MRB#262000-208).



- AMM 26-20-00-900-803: Discard of the Forward Baggage Compartment High Rate Discharge (HRD) Fire Bottle Cartridge (MRB#262000-211).
- AMM 26-20-00-900-804: Discard of the APU Fire Bottle Cartridge (MRB#262000-215).
- AMM 26-20-00-710-802: Operational Check of the Fire Bottle Pressure Switch Circuits on All Bottles (MRB#262000-217).
- AMM 26-20-00-710-803: Operational Check of the Aft Baggage Compartment Fire Extinguishing Circuits (CMR#262000-109).
- AMM 26-20-00-710-804: Operational Check of the Forward Baggage Compartment Fire Extinguishing Circuits (CMR#262000-112).
- AMM 26-20-00-710-805: Operational Check of the Aft Baggage Compartment Inlet and Outlet Vent Valves (MRB#262000-220).
- AMM 26-20-00-710-806: Operational Check of the Nacelle Fire Extinguishing Circuits (MRB#262000-204).
- AMM 26-20-00-710-807: Operational Check of the APU Fire Extinguishing Circuits (MRB#262000-216).
- AMM 26-20-00-720-801: Functional Check of the Lavatory Fire Extinguisher Bottle (MRB #262000-219).
- FIM 26-21-00-810-801: FIRE PROTECTION panel, an indication discrepancy of the BTL LOW light for the ENGINE 1 or ENGINE 2 - Fault Isolation.
- FIM 26-21-00-810-802: FIRE PROTECTION panel, an indication discrepancy of the FWD BTL light for the ENGINE 1 - Fault Isolation.
- FIM 26-21-00-810-803: FIRE PROTECTION panel, an indication discrepancy of the FWD BTL light for the ENGINE 2 - Fault Isolation.
- FIM 26-21-00-810-804: FIRE PROTECTION panel, an indication discrepancy of the EXTG AFT BTL light for the ENGINE 1- Fault Isolation.
- FIM 26-21-00-810-805: FIRE PROTECTION panel, an indication discrepancy of the EXTG AFT BTL light for the ENGINE 2- Fault Isolation.
- FIM 26-21-00-810-806: FIRE PROTECTION panel, an indication discrepancy of the PULL FUEL/HYD OFF light in the ENGINE 1 area - Fault Isolation.
- FIM 26-21-00-810-807: FIRE PROTECTION panel, an indication discrepancy of the PULL FUEL/HYD OFF light in the ENGINE 2 area - Fault Isolation.
- FIM 26-23-00-810-802: BTL LOW, APU (Caution) - Fault Isolation.
- FIM 26-23-00-810-803: FIRE PROTECTION panel, an indication discrepancy of the BTL LOW light in the APU area - Fault Isolation.
- FIM 26-23-00-810-804: FIRE PROTECTION panel, an indication discrepancy of the EXTG light in the APU area - Fault Isolation.
- FIM 26-23-00-810-805: FIRE PROTECTION panel, an indication discrepancy of the BTL ARM light in the APU area - Fault Isolation.
- FIM 26-23-00-810-806: FIRE PROTECTION panel, an indication discrepancy of the VALVE CLOSED light in the APU area - Fault Isolation.



- FIM 26-23-00-810-807: FIRE PROTECTION panel, an indication discrepancy of the FUEL OPEN light in the APU area - Fault Isolation.
- AMM 26-25-00-210-801: Visual Check of the Portable Fire Extinguishers Pressure Gauge (MRB# 262500-201).
- AMM 26-25-01-000-801: Removal of the Flight Compartment Fire Extinguishers.
- AMM 26-25-01-400-801: Installation of the Flight Compartment Fire Extinguishers.
- AMM 26-25-00-720-801: Functional Check of the Portable Fire Extinguishers (MRB# 262500-202).



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